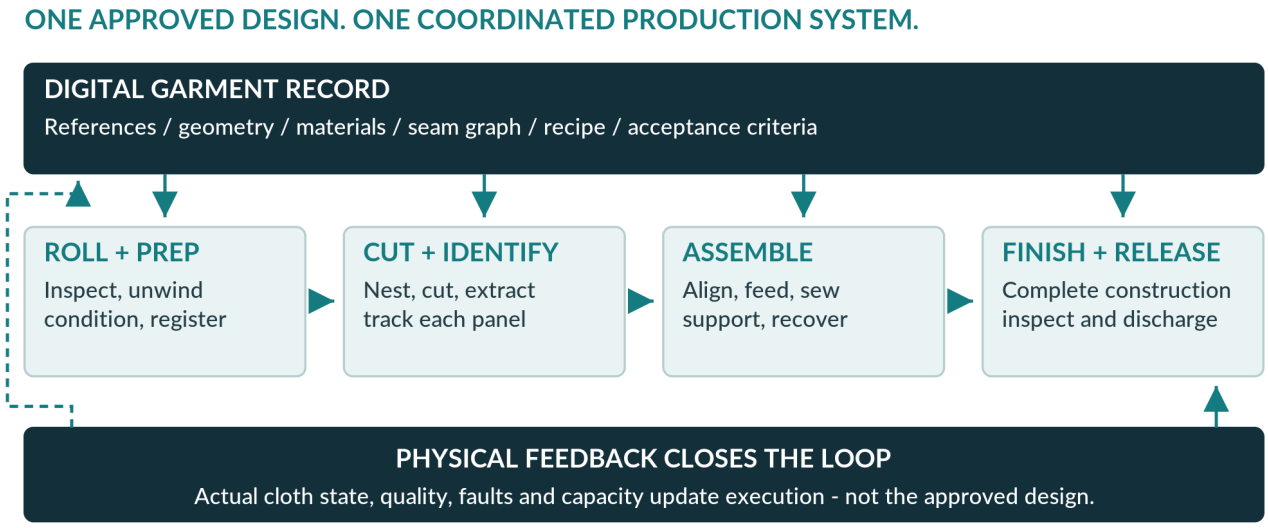
TECHNICAL + COMMERCIAL WHITE PAPER

AI-directed roll-to-garment manufacturing

A programmable industrial system for conventional fabrics, complete construction and flexible small-batch production.



Functional concept; no claim of a fabricated machine, verified footprint or rated output.

MECHANICS / AI / CAPABILITIES / INDUSTRY / ECONOMICS

Research snapshot: 19 October 2026 • Version 1.0

Research-grounded concept. Proposed functions and economic scenarios require engineering and commercial validation.

Contents / System and technology

Scope and evidence first; physical workflows and software follow the material through the cell.

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Reading routes

For the complete product: Executive synthesis, Sections 02–04 and 09. For controls and software: Sections 10–12 and Appendix B. For capacity and autonomy: Sections 13–16. For comparison and economics: Sections 17–22. For validation: Sections 23–24 and Appendices A–D.

Numbered citations link to the primary-source register. All original diagrams describe proposed functions, not fabricated hardware. Financial tables distinguish supplier contribution from garment-operator operating profit.

Executive synthesis

The proposed system is an **AI-directed roll-to-garment production cell**: a user develops and approves a garment in a conversational, visually grounded workspace, loads qualified fabric rolls and construction materials, and receives complete cut-and-sewn garments. The base process includes material preparation, cutting, component identification, robotic handling, sewing, required construction finishing, inspection and discharge. Embroidery, printing and other processes can be added as complete, integrated modules.

This is a product definition and engineering synthesis, not a description of an existing prototype. Its central hypothesis is that a common garment representation, reusable construction skills and mechanics designed for observable, recoverable handling can make a wide range of supported garments economical in smaller batches. The intended scope includes T-shirts, blouses, sweatshirts, hoodies and other tops, with longer garments added when their specific handling requirements are established.

The evidence supports taking that hypothesis seriously. Commercial equipment already addresses roll-fed single-layer cutting and individual sewing operations; vendor demonstrations connect cutter metadata to robotic component collection. Research demonstrates fixture-free sewing, controlled feeding of unequal fabric lengths, panel transport, CAD-assisted alignment and multimodal generation of parametric garment patterns. These are restricted capabilities with different starting conditions, not proof of a complete production cell. [01–10]

The hardest integration work is maintaining control of a garment as it changes from separate flat pieces into a connected three-dimensional object. A valid architecture must preserve or recover the intended edges, isolate the correct layers, support unused fabric, maintain sewing access, avoid trapping fixtures inside closed sections and release the finished item. AI can choose and correct actions only where the hardware provides the necessary reach, support and sensing.

The system is **not distinguished merely by using AI**. CreateMe explicitly describes mechanics and learned control developed together; Silana describes adaptive fabric handling and sewing; SoftWear promotes machine-vision garment worklines; and Resonance connects design intent to production orchestration. The proposed distinction is a combination to be demonstrated: conventional stitched construction, roll-to-output continuity, reusable capabilities across garment families, inexpensive job changes and a deployable product for smaller manufacturers. [14–19]

Product and performance objectives

Item	Proposed definition
Input	Qualified body, rib and other fabric rolls; thread, tape, labels, closures and specified consumables.
Output	Complete construction to an approved garment specification, plus an acceptance or reject record.
User interaction	Conversational design with reference uploads, pattern and garment views, technical documents, explicit production approval and live job status.
Base production objective	At least 100 acceptable premium T-shirts per 12-hour shift, measured from a disclosed loaded state.
Higher-capacity studies	200–300 shirts per shift through improved balance and handling; 500 as a separately costed configuration. These are scenarios, not ratings.
Base selling-price objective	Approximately US$100,000 for a defined delivered configuration. Cost feasibility is not established.

Item	Proposed definition
Autonomy boundary	No manual construction handoffs during a successful run. Setup, loading, maintenance and exceptional interventions are recorded separately.
Physical format	A modular industrial cell with guarded work zones, not a home appliance or a dimensionally validated cabinet.

What the economics show—and do not show

Two profit models are kept separate. For the **machine supplier**, illustrative build-and-test costs of $52,000, $83,000 and $140,000 produce very different results at a $100,000 selling price. After assumed delivery, warranty and variable selling costs, contribution is respectively $33,000, $2,000 and negative $55,000 per system. None of those build costs is a supplier quotation. The comparison shows how strongly the price objective depends on physical simplification and repeatable manufacturing.

For the **garment operator**, a worked premium blank T-shirt model assumes $6.08 in variable cost per acceptable garment and $293.86 in daily fixed operating cost including depreciation. At 100 acceptable garments sold per day for an assumed $12 ex-factory price, modeled operating profit is about $298 per day, or $74,415 across 250 operating days. At 50 sales per day the same model is approximately break-even. These are conditional arithmetic examples, not a margin forecast, market-price claim or guarantee of demand. Sections 20–22 expose every material assumption and the effects of lower utilization, higher labor and worse yield.

> The project's commercial test is not simply whether a robot can sew a shirt. It is whether one repeatable system can deliver approved quality, useful variety and competitive total cost across real small orders, without hiding labor or preparation outside the measured process.

01 / Purpose, scope and evidence

A research compilation, not a product certification

This white paper establishes a common technical and commercial definition for the proposed system. It brings the mechanics, software, garment requirements, production model, competitive landscape and unit economics into one document. All monetary amounts are in United States dollars unless otherwise stated; prices and rates in the numerical models are explicit assumptions rather than quoted market offers.

The research snapshot is **19 October 2026**. Public product pages can describe intentions as well as shipping functionality. Research abstracts and author pages demonstrate only the tasks they report. Where a source starts with already aligned pieces, a person loading a hemmer or chemically stiffened fabric, those conditions are retained in the analysis rather than silently removed.

Evidence language used throughout

Label	Meaning in this paper
Established process	An existing commercial process or equipment function, described within its actual operating scope.
Research demonstration	A reported experiment on specified materials, tasks and starting conditions; not a product warranty.
Vendor claim	A manufacturer's public description, not an independent audit of output, autonomy, economics or sustainability.
Proposed design	An architecture, workflow or requirement synthesized for this project; it must be engineered and tested.
Illustrative model	Transparent arithmetic using declared assumptions. It establishes a relationship between variables, not measured performance.

Schematics in this document are original **functional diagrams**. They are not dimensioned fabrication drawings, collision-checked layouts, licensed supplier designs or images of working hardware. A block labeled "collar station" names work that must be accomplished; it does not establish that the complete mechanism has been solved.

Terminology that prevents misleading comparisons

A **complete garment** has all construction operations in its approved specification finished. An **acceptable garment** also meets the agreed inspection criteria. A **production attempt** can end in acceptance, rejection or an interrupted cycle. A **recipe** is a versioned, qualified combination of design limits, material profile, tools, construction sequence and quality checks.

Autonomous assembly means no person manipulates the garment between the defined operations. **Unattended batch operation** additionally requires sufficient supplies, waste and output capacity, and acceptable handling of faults without intervention. An **automated changeover** is a separate capability: a machine can assemble a batch autonomously yet still require an operator to change rolls or thread before the next batch.

A **completion interval** is the time between outputs of a running pipeline. **Lead time** is the elapsed time for one garment to pass through the cell. **Accepted output** is not equivalent to needle cycles, completed hems or nominal robot speed. **Contribution** is revenue less specified variable costs; it is not company net profit. These distinctions are applied consistently to the performance and economic scenarios.

Scope of the conclusion

The synthesis identifies a credible engineering route based on existing process technology and demonstrated manipulation capabilities. It does not establish unrestricted garment support, a $100,000 delivered system, a particular footprint, specified shift output or commercially sufficient reliability. The development program must establish those properties together. A negative result on one architecture is evidence to redesign that architecture, not proof that intelligent garment manufacturing as a category is physically impossible.

02 / The product: one controlled material-to-garment process

The proposed product is a coordinated industrial system rather than a collection of independently operated machines. A single application manages the approved garment, production recipe, material inventory, cutting layout, component identity, station scheduling, inspection results and output record. Different modules can work on different garments simultaneously, but their handoffs share a defined control contract.

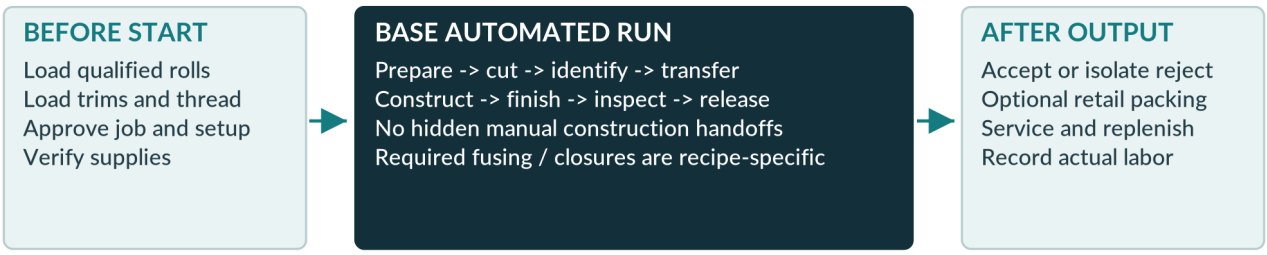


Figure 1. Base-system scope. The proposed unattended run excludes hidden manual construction; setup and retail handling remain explicitly visible.

What is inside the base process

The base system accepts qualified conventional textiles and construction materials, performs the necessary preparation, cuts the actual pattern components, extracts them without losing identity, assembles them and completes the required finishing. It must handle scrap, defective pieces and incomplete jobs rather than depend on an operator to sort them during a successful run.

The user can be required to load rolls into specified positions, thread or fit approved supply cartridges before Start, confirm material identity and approve a production job. A deliberate setup process is compatible with autonomous production. Conversely, repeated human placement of a neckline, manual fabric turning or a final hand-sewn closure is not autonomous assembly simply because the needle movement is automated.

Function	Base requirement	Boundary to disclose
Material preparation	Controlled unwind, registration, inspection and any preparation required by the supported recipe.	Qualified incoming dimensional stability or measured in-cell preparation; no concealed off-cell labor.

Function	Base requirement	Boundary to disclose
Pattern cutting	Layout, single-ply cutting, notches and necessary markings, cut verification and waste removal.	Maximum width, roll formats, tools and validated materials.
Component preparation	Extract and identify every required panel, rib piece and trim; support it for the next operation.	No externally pre-sewn collar or manually sorted kit presented as raw-roll input.
Assembly	Alignment, feeding, joining, trimming, transfer and required turning.	Permitted seam types, tool combinations, sizes and bulk.
Construction finishing	Hems, reinforcement, labels, closures and construction-required pressing/fusing.	A buttoned blouse requires a configuration that also completes its buttonholes and button attachment.
Inspection and discharge	Defined checks, reject isolation and a retrievable finished garment.	Destructive wash and seam tests qualify recipes; they are not performed on every saleable item.

What is outside the baseline

Fabric manufacture, yarn production, dyeing, special garment washes and broad chemical finishing are outside the baseline. Retail folding, packaging and shipping can be external or optional. A garment needing pressing or fusing to complete its actual construction cannot be advertised as complete while that step is excluded. Decoration modules have their own process requirements, utility limits and throughput ratings.

The term “any fabric” is not used. A released configuration must identify the material formats and properties it supports. Open-width jersey, woven fabric and ribbing are different handling cases; tubular ribbing, very slippery cloth, coated textiles or heavy seam stacks require explicit qualification. A target garment family is an intended capability, not evidence that every material and detail within that family is supported.

Operator experience

The intended journey is: create a project; upload references and specifications; select a qualified material; refine the garment; approve the technical record; load or verify supplies; authorize the batch; and retrieve accepted garments with a production record. The system should present a reasoned alternative when a request exceeds its capabilities before fabric is cut. It should never quietly simplify a neckline, replace a stitch with adhesive or omit a detail to make the physical task easier.

03 / Why the engineering direction is credible

The positive case comes from several independently developed capabilities. They reduce the number of physical principles that must be invented, but they do not eliminate the need for system integration. The table intentionally separates the evidence from the additional work required here.

Capability and source	Demonstrated or advertised scope	Additional requirement for this platform
Roll-fed cutting: Lectra Virga Fashion [01]	Commercial single-layer cutting with automatic roll-out and camera-assisted positioning.	Verified extraction, identity-preserving handoff and coordination with downstream assembly.

Capability and source	Demonstrated or advertised scope	Additional requirement for this platform
Cut-to-kit transfer: RobotFactory RPT 130 [02]	Cutter metadata supports picking and sorting T-shirt components. Its example sends them to a sewing department.	Transfer in a controlled pose directly into autonomous garment construction.
Fixture-free sewing: Tang et al. [03]	Two manipulators guide prepared, aligned fabric pairs through an ordinary sewing machine using learned edge detection and layered control.	Seam preparation, all connected-garment handoffs and complete finishing.
Unequal-length feed: Jin et al. [04]	Independently driven rollers sew stretchable layers of different lengths along a straight line; leading ends begin positioned at the foot.	Automatic loop formation, loading, curved neckline handling and release.
Supported panel transport: RollUP [05]	Motorized rollers and suction roll up fabric; coordinated units manage tension during transport.	Compatibility with the intended materials and with already-sewn assemblies.
CAD-assisted alignment: Dong et al. [06]	Observed fabric and known geometry align edges and sewing lines, including partial occlusion; starts with flat, wrinkle-free panels.	Recovery from wrinkles, folds and topology changes in a growing garment.
Edge recovery: Sunil et al. [07]	Vision and touch support edge grasping and sliding toward corners in deformable cloth.	Reliable manufacturing-specific skills with controlled quality outcomes.
Pattern representation: GarmentCode and Design2GarmentCode [08,09]	Parametric pattern programs and multimodal design-to-pattern generation.	Physical manufacturability, process planning, material calibration and machine qualification.
Tubular finishing: JUKI bottom hemmer [10]	Automated hemming and stacking after an operator loads the garment.	Automatic loading, orientation, release and integration with the remaining operations.

The integration problem is state continuity

Each module must receive something it can actually work with. A cutting system may know a piece's location at the instant of cutting, but extraction changes its shape. A sewing head may accept two aligned layers, but the previous operation may deliver a partially closed garment with the relevant edge underneath another panel. A hemmer may sew a tube reliably after loading, yet loading that tube is itself a manipulation task.

The proposed platform therefore treats each transition as a first-class operation. It must know which object is present, where the next edge should be, which supports currently control it, which layers belong under the needle and whether the receiving station is ready. A motion command is not sufficient evidence that the intended transfer happened.

Why an industrial cell is a better initial physical format

Dedicated work zones allow permanently threaded sewing functions, accessible fixtures, material buffers and serviceable modules. A single-purpose industrial head can be retained while its loading and unloading are automated. This avoids requiring every process to be miniaturized and combined in one compact enclosure at the same time that the handling method is being developed.

That is an architectural advantage, not a claim that the proposed price is easy. More stations can increase cost, floor area, calibration effort and shared-transport demand. A larger cell still needs a

complete garment motion study, a timing model and a service plan. The evidence justifies investigating the system; it does not permit adding nominal component capacities together and calling the sum a garment-production rate.

04 / Physical architecture and module interfaces

The starting architecture is a **hybrid modular cell**. Predictable operations use simple supports and fixtures where those reduce complexity; variable operations use active grippers and sensor feedback. Neither “no fixtures” nor “maximum robots” is a requirement. The objective is controlled fabric access at acceptable cost, with as little garment-specific rebuilding as practicable.

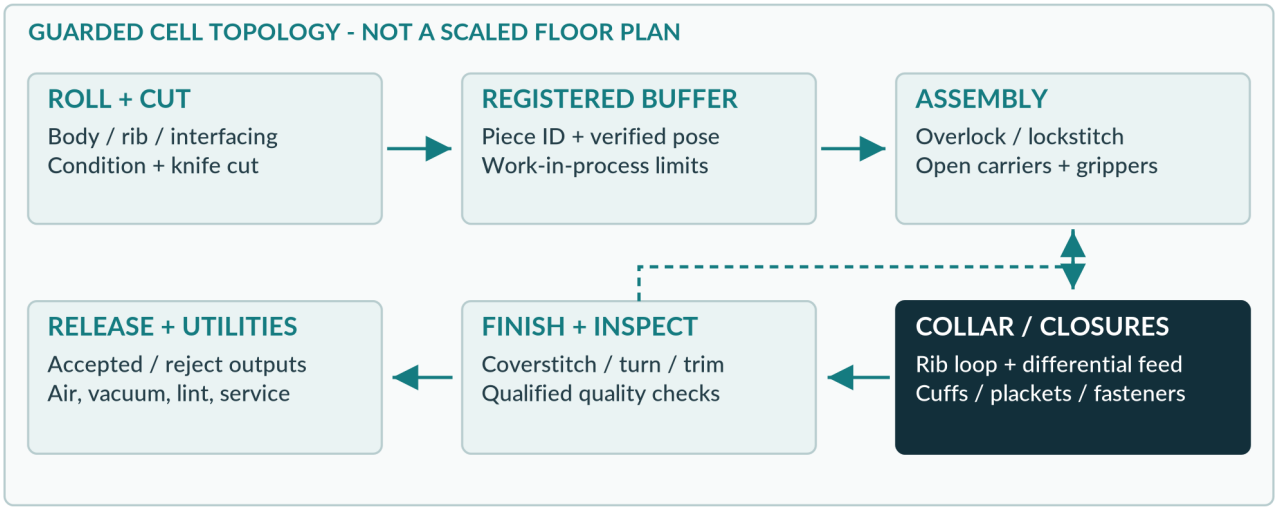


Figure 2. Relational layout of the proposed cell. Module positions convey workflow, not dimensions or a verified robot reach envelope.

Functional zones

The front end contains roll loading, controlled feed, inspection, conditioning where required and a supported cutting area. A registered buffer holds identified parts and growing garments. Construction modules provide structural joining, rib and neckline operations, hems, closures and required forming or fusing. An output zone separates accepted garments, rejects and recoverable work. Utilities include controls, guarding, air or vacuum generation or declared plant connections, lint extraction, waste collection and maintenance access.

A gantry can provide repeatable planar coverage while an articulated manipulator offers lifting and reorientation. Opening carriers can retain panel identity and support selected regions. The appropriate combination depends on the largest parts, seam access, material response, payload with tooling, transfer time and the ability to release a closed garment. Selection should follow real-fabric trials and kinematic analysis, not the visual appeal of a robot arm.

Every handoff needs a contract

Interface field	Required meaning
Identity	Job, garment, panel or subassembly, material lot, recipe revision and tool configuration.

Interface field	Required meaning
Geometric state	Actual observed pose or supported boundary, face and grain orientation, known seam locations and uncertainty.
Support state	Which carrier, gripper or surface currently holds each relevant region; permitted release order.
Material limits	Maximum permitted tension, contact force, stretch or temperature for the operation.
Readiness	Receiving station available, correct tooling present, supplies sufficient and safe state confirmed.
Acceptance	Sensor-confirmed layer count, grasp, edge access and transfer success.
Failure response	Retry limit, permitted recovery, reject route, recut request and safe-stop conditions.

The receiving module acknowledges control before the source releases fabric. The controller records a transition from one verified state to the next, not merely “robot move completed.” A failed acknowledgment should not leave a garment unsupported between machines.

Space, mass and utilities

No exact footprint or weight is assigned at this stage. A layout must include roll diameter and mass, maximum fabric width, cutting travel, panel swept volumes, tool reach, output accumulation, guarding and maintenance clearance. Holding a panel does not establish that the joined garment can rotate or escape the same workspace. Additional floor area may be required for loaded rolls and safe loading equipment even if they are outside the guarded moving zone.

Utilities must be measured for the complete system. Sewing motors, extraction, compressed air, vacuum hold-down and compute draw power differently; printing or fusing can add heat and ventilation requirements. Shared plant air is not “free” simply because it is absent from the machine cabinet. The supplier specification should identify peak demand, average duty, heat rejection, air quality and service isolation requirements.

Buy, adapt or build

Established process modules can accelerate a prototype, while purpose-designed mechanisms may be necessary for a viable repeat-build cost. Each outsourced module needs an accessible control interface, fault reporting, maintenance documentation and a defined support boundary. Research availability is not permission to reproduce patented hardware or commercially redistribute code. Supplier integration rights and licensing require separate review; no clearance determination is made here.

05 / Materials, roll handling and dimensional stability

“Load a roll” is an input format, not proof that the material is ready for accurate manufacture. Oshima’s preparation documentation identifies retained tension from production, finishing, storage and transport as a reason to relax fabric before cutting. The necessary method and duration depend on the material. A fast in-cell feeder cannot be assumed to replace all conditioning. [11]

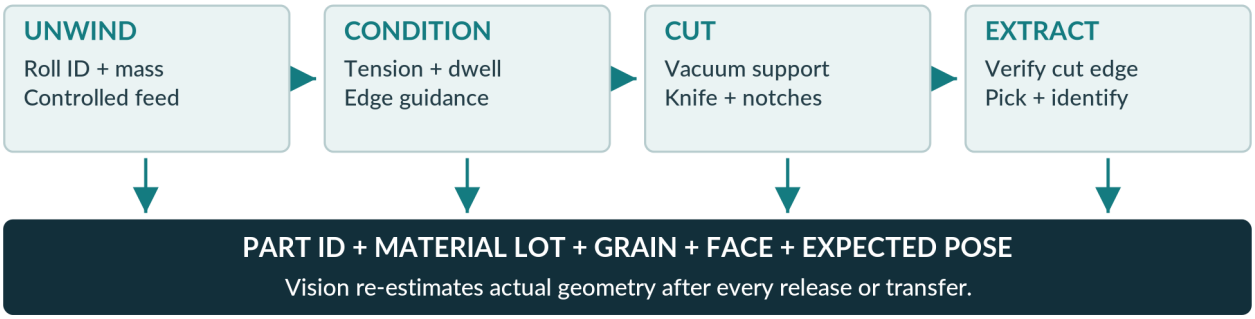


Figure 3. Proposed roll-to-panel front end. Material identity is preserved while actual position and geometry are measured again after cutting.

A qualified material profile

The material record should include composition where known, construction, mass per area, thickness, usable width, lot, face and grain directions, stretch under a defined test load, recovery, surface friction, edge curl, porosity, shrinkage after the agreed wash protocol and sensitivity to pressure or heat. These are proposed qualification fields, not values inferred confidently from a photograph or fabric trade name.

For each profile the system needs acceptable ranges and handling settings. “Cotton jersey, 280 gsm” is not a complete robotics specification. Two materials carrying that description can require different grip pressure, tension, blade settings or seam feed. A new supplier lot may be accepted through a defined verification procedure or require a new profile rather than silent reuse of the old one.

Controlled material delivery

A candidate feed path uses a roll cradle, controlled unwind, tension measurement, edge guidance and an accumulation section before the cutting surface. The system must detect end-of-roll, skew, folds, unusable width and relevant visible defects. A supported material may require a longer preparation buffer or an explicitly stabilized incoming roll. The stated production time must include whatever preparation the cell actually performs.

The cutter should receive stable fabric without pulling it into an incorrect shape. Low tension is not the same as uncontrolled slack: excessive slack can introduce wrinkles and placement errors. Closed-loop feed should keep the material inside its qualified handling envelope. The control policy must be tested when roll diameter changes and when a fresh roll is loaded, not only at one convenient state.

Multiple materials and colors

A shirt may use body fabric, rib, tape and labels from different supplies. A woven blouse may add interfacing and closures. The system therefore needs separate input identities and synchronization. A production job reserves all required components before irreversible work begins, reducing the chance of cutting a body that cannot be completed because its matching rib is unavailable.

Automatic switching between rolls, thread colors and trim cartridges is an additional mechanical capability. A basic configuration can require disclosed setup before a compatible batch while still being autonomous during that batch. The software should expose those restrictions when estimating a mixed-order schedule. A user selecting a new color on screen does not establish that a roll and every matching thread path have changed themselves.

Consumption and batch capacity

The model must count input material used for nested shapes, cut waste, failed parts, quality rejects, setup pieces and rework. Remaining roll length should be tracked with uncertainty and checked against the next batch. A system intended to operate unattended also needs sufficient rib, thread, tape, label and output capacity. End-of-supply detection is necessary but does not by itself provide automatic replenishment.

Section 21 gives a mass-balanced example. Under its assumptions, 100 acceptable heavyweight T-shirts consume approximately 43.2 kg of body and rib fabric including layout waste and rejects. That is a model result, not a general material requirement; actual pattern areas, weights and yields determine the number. Roll logistics must be sized to those actual requirements rather than the headline count of garments.

06 / Cutting, nesting, extraction and scrap

The proposed starting process is supported **single-layer knife cutting**, chosen for study because it can place different sizes and approved designs in a shared layout. Multi-ply cutting may be attractive for larger repeated orders, but it introduces stack registration and singulation requirements. The architecture should compare both against the intended order mix rather than declare either universally superior.

Lectra advertises automated roll-out, camera-assisted positioning and a 120 m/min cutting-path speed for Virga Fashion. That number is neither garment throughput nor fabric-roll consumption. Notches, curves, acceleration, feed, material support and offloading all contribute to cycle time. RobotFactory's textile application demonstrates the distinct next step: using cutter metadata to collect and sort identified T-shirt pieces before they go to sewing. [01,02]

Nesting is constrained optimization

A marker must respect grain direction, stretch direction, face, nap, print placement, pattern matching, usable width, known defects and seam allowances. A layout with high area utilization is not acceptable if it rotates a sleeve against the approved grain. A cutting optimizer also needs to consider downstream demand: cutting many panels for blocked stations can create excessive work in process and bury components required for urgent jobs.

Each cut component receives a digital identity linked to its garment and material lot. That identity need not require a permanent visible mark. Known cutting location, temporary machine-readable labels where approved, controlled carriers and verified transfer can maintain identity. Physical labeling decisions must consider garment appearance, adhesive residue, washing and any later cutting or sewing operation.

Extraction is not just picking something up

After cutting, the system must determine whether the perimeter is complete, separate the panel from surrounding waste and lift it without excessive distortion. Small notches, bridges, curled edges and adjacent pieces can complicate pickup. A panel-shaped vacuum pattern may help on some materials but leak or lift unwanted layers on others. A pinch or hybrid gripper requires safe contact regions and reliable edge access.

Scrap removal is coordinated with pickup. Pulling away the unused web must not drag a completed piece into the waste stream or move remaining components beyond recognition. The production controller should track extraction outcomes and distinguish an absent panel from a panel that was picked and misidentified. A recut must use a new part identity tied to the same garment need, so the rejected original cannot accidentally re-enter assembly.

What the cutter hands to assembly

The key output is not a bin of fabric. It is a **verified component in a usable supported state**, with identity, orientation, actual geometry, known defects and a clear next operation. This is a proposed extension beyond cutter-linked sorting. It reduces ambiguity at the assembly entry point while acknowledging that geometry changes whenever cloth is lifted, folded or released.

A buffer should store components in a way that preserves access and avoids flattening one garment under many unrelated kits. There may be separate buffers for flat pieces and sewn subassemblies. Their sizes follow the timing model and fault behavior, not an arbitrary number of trays. A full buffer must cause controlled upstream slowing or replanning rather than physical overflow.

Cutting-related validation

Acceptance tests should cover largest and smallest parts, narrow rib strips, notches, local curves, roll skew, empty areas, varying porosity, visible defects and incomplete-cut recovery. Record geometric error after release, extraction success, material damage, lost identity, recut workload and scrap fraction. A cutter capable of accurate flat patterns is useful evidence; autonomous transfer in the condition needed for sewing is the additional system milestone.

07 / Manipulation: controlling cloth rather than commanding coordinates

The proposed controller operates on the **observed cloth state**. It uses the known garment and completed-seam record to narrow the task, then cameras and contact feedback to establish where the material actually is. A fixed coordinate can be a starting estimate; it is not a substitute for checking that the intended edge is present and the correct layers are controlled.

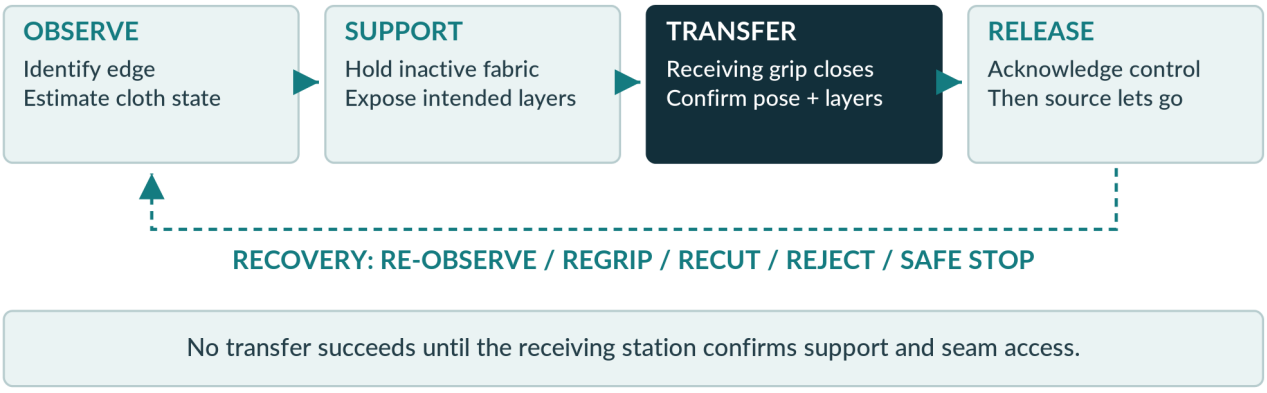


Figure 4. Proposed verified handoff. The receiving grip, support and seam access are checked before source release.

Selection of handling mechanisms

Mechanism to evaluate	Potential role	Main limitation to test
Pinch gripper with local compliance	Isolate an edge, regrip or lift a panel.	Double layers, fabric marking, difficult access and slipping.
Vacuum or distributed suction support	Hold broad areas and stabilize pieces on a surface.	Porosity, wrinkles, leakage, energy use and unintended pickup.

Mechanism to evaluate	Potential role	Main limitation to test
Rolling support	Reduce unsupported panel length during transfer.	Unrolling accuracy, edge curl and incompatibility with bulky sewn regions.
Opening or segmented carrier	Preserve registration and expose selected seams.	Access after topology changes; avoidance of trapped fixtures.
Driven edge rollers	Meter and align layers near a sewing head.	Limits in local stretch, curvature, contact force and lead-end introduction.
Movable garment cradle	Support inactive fabric during joining and turning.	Entanglement, transfer delay and reliable separation of layers.

This is a comparison agenda, not a selected bill of materials. RollUP provides research precedent for rolling and suction-assisted transport, while visuotactile work supports edge interaction under partial occlusion. Neither supplies an unrestricted gripper for the proposed garment range. [05,07]

Recovering the next edge

A folded edge is not automatically a terminal failure. A qualified skill can observe, lift a known region, expose an edge, grasp it and verify the result. But the mechanism needs the necessary degrees of freedom, free space and support. If a carrier blocks access or a garment is trapped around a solid support, the solution requires a different fixture or assembly order, not merely a more confident model prediction.

The controller should separate uncertainty about identity from uncertainty about pose. It may know which panel is present but not exactly where its edge lies. It may instead see a plausible edge but be unable to confirm whether it is a hem, a folded layer or a matching seam. Those conditions warrant different probes and stop thresholds. Force or tactile checks should be used where they add discrimination rather than treated as decorative sensors.

Entry conditions and bounded skills

A reusable skill has a required starting state, tools, observations, adjustable parameters, success test and permitted recovery path. For example, “align and sew” requires known mating edges, controlled layer count, an accessible lead-in, adequate thread and support for unused cloth. It must not begin simply because the next line in a program says “sew.”

A skill library can cover exposure, pickup, flattening, tensioning, alignment, feeding, folding, turning, transfer and release. Generality comes from parameters and observed state, but only within an evaluated envelope. New garment sizes and materials should be tested against those limits. A novel combination of familiar skills can still fail because one operation leaves the next without access or adequate support.

Learning and physical data

Training records should include unsuccessful grasps, doubles, slip, occlusion, malformed seams and recovery attempts, not only ideal runs. Physical data must connect sensor observations to action, material, geometry and outcome. Simulation can help explore reach, scheduling and candidate policies, but real textile response and seam quality require physical testing. A learned policy is released as a versioned component; improvements are evaluated offline before replacing a qualified production version.

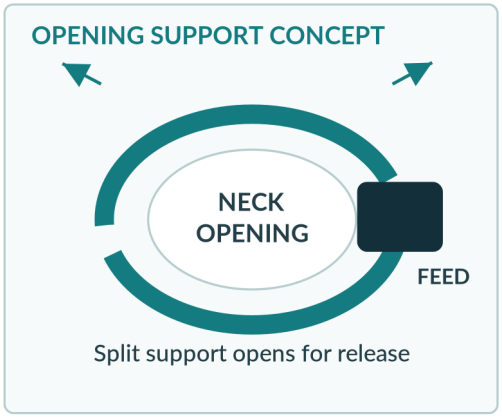
08 / Stitching, ribbing, closures and construction tools

The base system preserves conventional cut-and-sew construction. Its capability derives from the combination of stitch-forming mechanisms and autonomous material presentation. A sewing head's ability to create a stitch does not establish that the robot can introduce the correct layers, follow the seam, avoid neighboring cloth, finish the thread path and remove the garment.

A proposed process toolbox

Structural knit seams call for appropriate overlock capability; visible knit hems call for coverstitch capability; many woven assemblies and topstitch details call for lockstitch capability. Those functions can be implemented as separate permanently threaded modules in the initial industrial architecture. Buttonholes, button attachment, bartacks, selected zipper operations and fusing require their own qualified tools or modules where included in a released garment configuration.

Automatic needle or thread conversion is not presumed. Multiple heads may simplify process changes but add space, thread paths, maintenance and transport. Automatic thread trimming and break detection do not eliminate all thread faults. Any necessary bobbin, looper-thread, needle or tool intervention is either automated and validated, completed before the run or included as an interruption.



Candidate mechanism only; support geometry, circular handling and release require real-fabric tests.

Figure 5. Split-support ribbed-neckline concept. A possible support-and-release arrangement; its geometry and full loop handling are unproven.

The ribbed collar as an integrated benchmark

A complete collar module must cut the strip from the designated rib supply, join its ends where required, establish face and orientation, form the fold, present it to the neckline, distribute local stretch, sew the approved seam, perform tape or topstitch steps and release the garment. It cannot receive a manually prepared loop while being described as roll-to-garment automation.

The differential-feed research demonstrates controlled sewing of unequal-length stretchable layers, but its prepared leading ends and straight seam are narrower than this complete task. It supports studying an independent feed mechanism; it does not validate automatic circular installation. [04]

For a numerical illustration, a 50 cm body neckline with a 42.5 cm relaxed rib seam line has a nominal ratio of 0.85. That is not a universal rib rule. Rib recovery, stretch force, desired appearance, seam allowance and the actual material pair determine the approved relationship. The system must control local distribution rather than simply stretch the entire loop to a target length once.

A proposed opening support would retract or split after sewing. It must leave the rest of the shirt supported while the neckline is unloaded. Cameras and contact checks should distinguish a correctly placed loop from a twist or an unwanted captured layer. A collar looking symmetrical under tension is not enough; geometry, recovery and seam appearance should be checked after release and during wash qualification.

Turning, fusing and closures

A woven collar or cuff may require shaped joining, trimming, turning, pressing and topstitching. Those are different manipulations from feeding a flat seam. Facings and interfacing introduce additional layers and material compatibility. A module must provide a controlled way to apply and release heat or pressure without damaging the garment or creating a fire or entanglement hazard.

Closures require defined supply formats and presentation. A button must be oriented, placed and attached; its corresponding buttonhole must exist and be verified. A zipper requires accurate tape placement, allowance management and the specified end treatment. A drawcord operation involves feeding, locating ends and verifying the result. These operations should appear in the recipe and costing rather than disappear into a generic “attach trims” label.

Finishing equivalence matters

An open-flat hem before side closure can differ at the seam intersection from a specified tubular hem completed afterward. A simpler neckline construction can differ from a folded rib collar. The compiler may propose a more machine-friendly method, but the customer must approve any resulting construction change. Software optimization is constrained by the actual garment specification, not merely by an approximate silhouette.

09 / Complete garment workflows and product breadth

The platform should be validated through complete garment recipes, not a collage of successful operations on unrelated test pieces. The first premium T-shirt gives a specific integration target; woven and bulky-rib tasks should challenge the same architecture early enough to prevent a permanently single-purpose design.

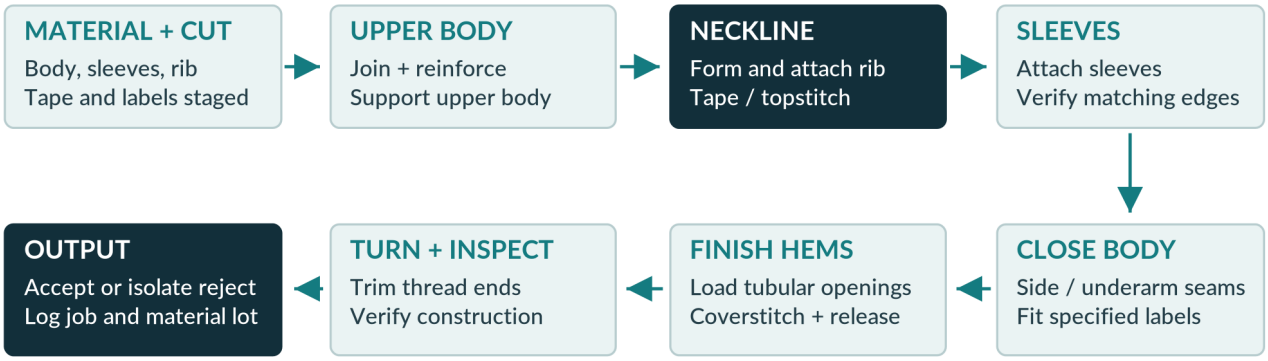


Figure 6. Candidate premium T-shirt route. Exact seam order is open to engineering, but every dependency and finishing requirement must be preserved.

Premium heavyweight T-shirt reference recipe

The initial sample specification should freeze a particular cotton jersey in the proposed 240–300 gsm range, a matched rib, shoulder or neck reinforcement, structural seams, sleeve and body hems, labels and approved dimensions. This range defines an intended test category, not identical behavior across all textiles of that weight.

Stage	Required operations	Completion evidence
Prepare materials	Verify body and rib lots; provide required relaxation; reserve tape, labels and thread.	Qualified material state and enough supply for the disclosed batch.
Cut and identify	Cut front, back, sleeves and rib; extract pieces; manage scrap; associate trims.	All components identified, oriented and held in usable states.
Build upper body	Align shoulders; apply specified reinforcement; join with the approved seam.	Correct seam allowance, layer count and reinforcement location.
Construct neckline	Form rib loop and fold; introduce to neckline; distribute feed; sew and complete agreed tape/topstitch.	Untwisted, released collar with correct geometry and no caught body layer.
Attach sleeves	Match sleeve and body edges, controlling curvature and any permitted easing.	Approved sleeve orientation, seam and attachment shape.
Close body	Join side and underarm paths; include seam-integrated labels where specified.	Correct layers, unblocked openings and complete seams.

Stage	Required operations	Completion evidence
Finish hems	Load sleeve and body openings onto suitable supports; fold and coverstitch; finish joins.	Consistent hem width and approved seam-intersection construction.
Complete and release	Trim threads, perform necessary turning, inspect and discharge.	Wearable garment meeting the agreed checks or an identified reject.

The reference route is not a claim that every step must occur in that order. A revised sequence is acceptable when the physical construction and quality remain equivalent. All changes need a support-and-access analysis: what holds the garment, which edge is reachable, how unintended layers remain clear, and how the item leaves the tool.

Woven blouse reference family

A defined blouse recipe starts from a selected woven material, the relevant interfacing and closures. Candidate preparation includes darts or shaping seams, placket formation, facing or collar subassemblies, necessary fusing, shoulder and sleeve construction, side closure, cuffs, hems, buttonholes, buttons and turning or pressing. The exact list depends on the approved design; no garment should inherit unnecessary operations merely because another blouse uses them.

This family stresses precise folding, small components, layer stacking, turning corners and closure positioning. A new visual design may reuse an approved placket and collar; a genuinely different collar construction may require a new tool or skill. Lightweight transparent, slippery or bias-cut fabrics need their own material qualification rather than being treated as ordinary stable woven panels.

Sweatshirts, hoodies and longer garments

Sweatshirts add bulk, thick seam crossings and rib cuffs or waistbands. Hoodies add hood assembly and attachment, pockets and optional cords or zippers. A pullover hoodie without a zipper is a different process configuration from a lined zip-front style. A machine's ability to attach a rib neck does not by itself establish that it can feed and release a much smaller cuff opening at the required seam thickness.

Sweatpants introduce long panels, crotch geometry, pockets, leg tubes and waist elastic or drawcord handling. The common platform may reuse skills, but reach, support and access must be demonstrated on those actual assemblies. Longer parts do not necessarily require a flat bed equal to their full length, yet any folded or rolled support strategy becomes part of the validated mechanism.

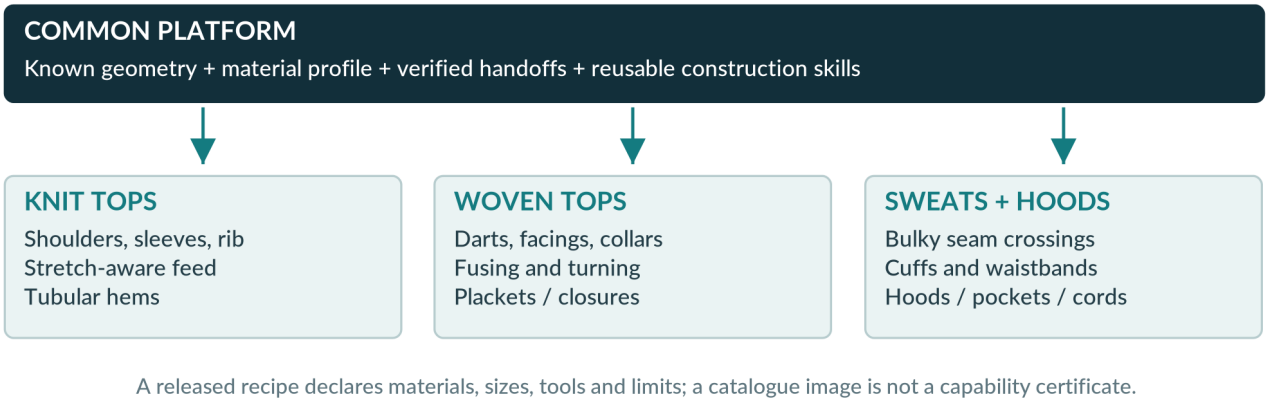


Figure 7. Versatility through construction families. Common handling and information structures support different tool and recipe combinations.

Released capability matrix

Family	Intended common capabilities	Additional configuration or qualification
T-shirts and related knit tops	Knit structural seams, sleeve attachment, rib neck, hems and labels.	Neckline variants, size range, fabric response and sleeve-length handling.
Sweatshirts	Shared knit operations and rib feeding.	Cuffs, waistbands and maximum seam-stack thickness.
Hoodies	Shared body assembly and selected pocket skills.	Hood geometry, cord channels, closures or lining where specified.
Woven blouses	Cutting, identity, manipulation and lockstitch workflows.	Darts, fusing, shaped collars/cuffs, plackets, buttons and buttonholes.
Sweatpants and similar bottoms	Shared feeding, hems, labels and inspection.	Longer-piece support, crotch access, leg turning and elastic installation.
Decorative variants	Approved base garment plus routing.	Complete print or embroidery module and a separate production rating.

A useful breadth test is a withheld, supported design variation executed through existing skills without manually programming each trajectory. A stronger test spans two construction families on the same core handling system. The final report must distinguish changed parameters, changed tooling, new software and manual setup. “Same machine” should not hide replacement of most of the mechanism between demonstrations.

10 / The conversational design and production workspace

The user interface is the primary entry point, but it is not the entire intelligence of the system. Its purpose is to turn intent into a consistent approved garment, expose manufacturing constraints early and preserve the connection between design changes and physical production. It should be chat-led and visually grounded rather than a text-only sequence of prompts.

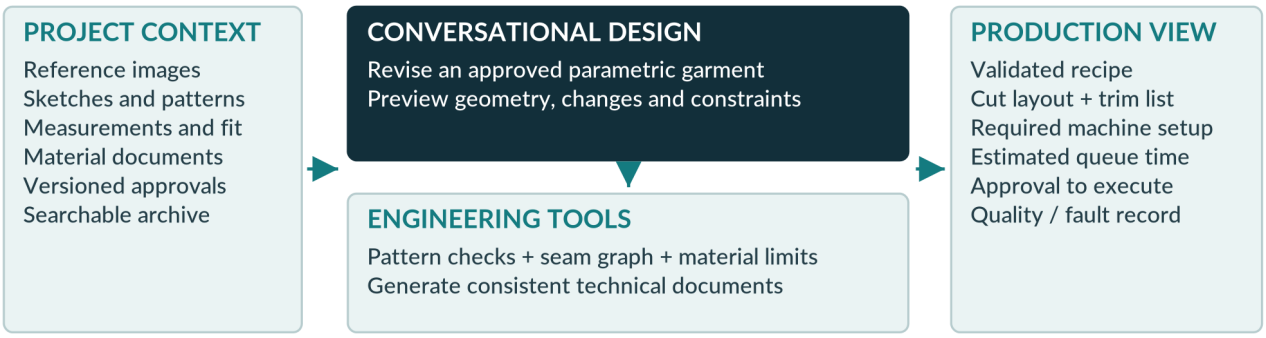


Figure 8. Proposed application workspace: context, conversational revision, engineering tools and production status linked to one approved record.

Persistent projects and reference handling

A project can contain photographs, sketches, pattern files, measurement tables, fabric documentation, prior technical packs, sample feedback and production results. The reference archive is searchable and persistent; the currently approved garment is a separate structured record. Older images should not silently override an explicit later decision. Each extracted fact should retain its source and confidence where uncertainty affects manufacture.

A photograph can suggest a silhouette or detail but does not establish an exact dimension, fabric composition or tested stretch profile. The application must request missing measurements or label estimates before approval. Extensive stored context does not imply unlimited active model context or perfect memory. Relevant retrieval, structured decisions and version history are the proposed means of managing large projects reliably.

Interactive design functions

The workspace should display the garment, pattern geometry and technical specification together. A user might select a sleeve and request a two-centimeter length change, compare necklines, specify a rib width or choose between approved construction alternatives. The change should update every dependent output, not only a rendering. GarmentCode and Design2GarmentCode provide evidence for parameterized pattern construction and multimodal pattern-program generation; the physical production checks remain an additional layer. [08,09]

The interface distinguishes a **concept preview**, an **engineering-checked design**, a **physically qualified recipe** and an **approved production job**. These are different states. A visually convincing garment can have mismatched seam lengths or require a tool the cell lacks. A physically qualified base recipe can become unqualified after a significant material or construction change.

Technical documentation generated from the record

Proposed outputs include a versioned technical pack, measurement and tolerance table, panel list, material and trim specification, grain/face annotations, seam map, stitch requirements, cutting marker, construction sequence, labeling instructions, machine configuration and inspection plan. Documents are generated from the same source data used by the execution system. A rendering or narrative description is not a substitute for actual cutting geometry.

Before production, the app should show estimated material consumption, required input rolls, machine/tool compatibility, expected setup, order quantity and the assumptions underlying time and cost. A visible approval action authorizes a fixed revision. Later changes create a new revision and cannot silently modify jobs already in progress.

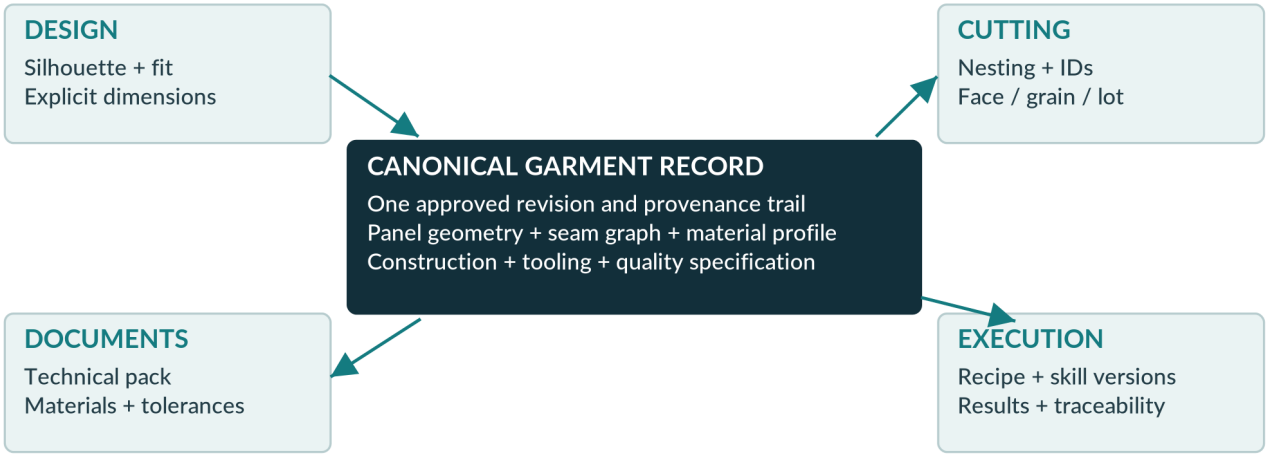
Production and exception experience

The user should see jobs waiting, cutting, assembled, inspected, accepted or rejected, with reason codes for interruptions. Operators need actionable instructions for authorized setup and service, not opaque model output. A remote person guiding a robot through a failure is still an intervention and must be recorded as such.

Where multiple customers share a production cell, access controls separate their design files, measurements and outputs. Reference permissions, retention choices and deletion requirements should be explicit. A customer's private design must not automatically become training data for other users. Production telemetry can be useful without retaining every identifiable reference or body-measurement file indefinitely.

11 / The garment model and design-to-machine compiler

The proposed **canonical garment record** is the engineering bridge between an expressive interface and controlled manufacture. It specifies what the garment is, how its pieces relate and which verified processes can construct it. The record is not a single image or a long prompt; it is structured geometry, topology, material information, decisions and constraints.



A material, neckline or seam change invalidates dependent outputs until they are rechecked.

Figure 9. A shared garment record drives the design, cutting files, technical documents and execution state. Dependencies are revalidated after a change.

Proposed record structure

Record group	Representative fields
Identity and provenance	Project, garment family, revision, approval, creator permissions and referenced evidence.
Geometry	Parametric panel boundaries, seam lines, allowances, notches, internal markings, darts and dimensions.
Material	Roll/lot, profile revision, face, grain/stretch direction, thickness, permitted tension and conditioning state.
Topology	Which edges join, their correspondence, seam type, orientation, easing or differential-feed relationship.
Trims and details	Labels, tape, interfacing, closures, reinforcement, decoration and exact attachment requirements.
Construction	Dependency graph, proposed sequence, intermediate garment states and support/release requirements.
Capability	Required tool and skill versions, size/material limits, known exclusions and setup requirements.
Quality	Measurement tolerances, seam criteria, approved sample, recipe qualification and inspection rules.
Production	Cut-piece IDs, station assignments, observations, actions, faults, material use and accepted/rejected outputs.

A seam graph connects specific panel edges rather than assuming that two nearby visible boundaries should be sewn. Edge correspondence can include notches and local easing constraints. Different seam lengths are not automatically errors: intentional rib stretch or easing must be represented explicitly. Unintentional mismatch must not be “fixed” by the robot stretching a layer until it fits.

Compilation and validation stages

The compiler first checks geometry, complete panel definitions, allowances and edge correspondences. It then checks material and tool compatibility, generates a candidate assembly dependency graph and evaluates whether each step can be supported, reached and released. Station allocation and cutting preparation follow only when the recipe can be executed on the actual cell configuration.

Physical access is the crucial extra check beyond pattern validity. A proposed seam order may close a tube before a required label or cuff is installed, trap a carrier, hide an edge from every gripper or require a turn the mechanism cannot perform. The compiler should search permitted alternative sequences and explain unsupported requests. It cannot certify real-cloth behavior solely from geometric simulation.

Change propagation

A neckline edit changes the panel edge, rib geometry, length relationship, cutting layout and construction parameters. A material substitution may affect shrinkage compensation, gripping, feed, pressing and quality qualification even when the silhouette is unchanged. A closure change can alter both the part count and assembly order. The system should invalidate dependent outputs until the relevant checks pass again.

Software-only variation is a desired outcome, not an assumption. Within an approved parameter range, a new size might reuse tools and skills. Outside that range, the machine may need a different support or qualification. The application must make that boundary visible to the user before an order is promised.

Executable jobs and module contracts

A job package should include signed or otherwise integrity-protected recipe and policy versions, material identities, station requirements, entry conditions, allowed parameter ranges, recovery limits, quality checks and a log schema. Each module accepts only instructions compatible with its capabilities and current state. Type-checked interfaces reduce ambiguity but do not replace physical validation.

The compiler and scheduler can use search, rules, optimization and learned models where appropriate. The most valuable output is a verifiable plan, not the presence of an LLM in every software layer. A repeat order should reuse an approved recipe while adapting to measured cloth conditions rather than re-invent its construction from free-form text.

12 / AI, robotics and deterministic control

The system is designed around **adaptive physical execution**, not a chatbot directly driving motors. Learned models can interpret references, estimate cloth state, select manipulation skills and predict outcomes. Deterministic software can enforce geometry, interfaces and scheduling constraints. Local controllers execute synchronized motion, while independent safety functions can inhibit it regardless of model output.

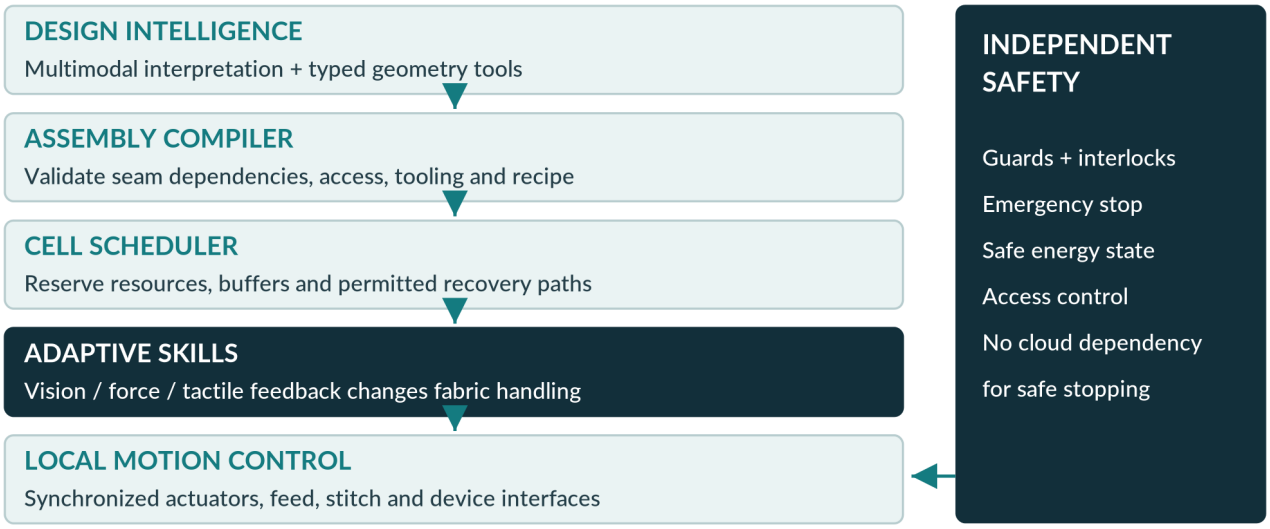


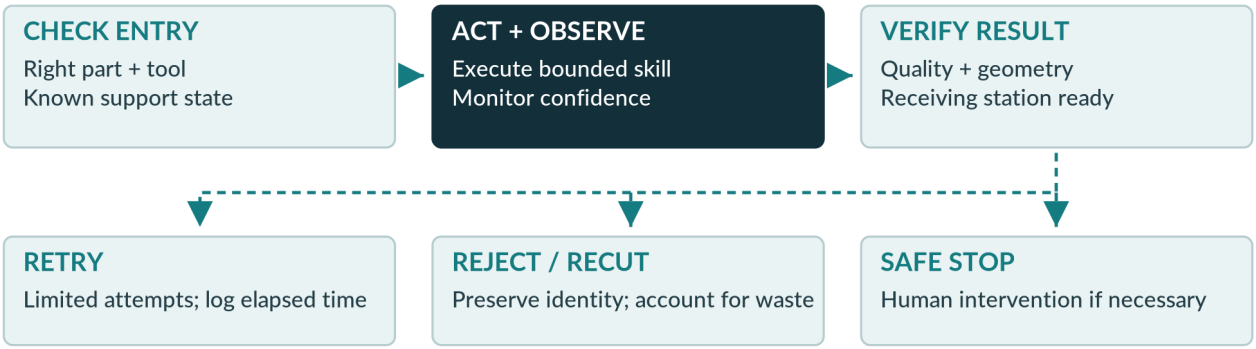
Figure 10. Proposed control stack. The safety path is independent of conversational inference and cloud connectivity.

What “designed for AI” should mean

The hardware should make relevant material states observable and recoverable: cameras need useful sight lines; grippers need ways to expose edges; supports need controlled release; and tools need measurable entry conditions. Mechanical simplification and sensing can reduce the uncertainty that a learned policy must handle. The goal is not to remove all structure and demand that a general model solve an arbitrary pile of cloth.

This principle is not unique to the project. The competitive comparison later describes other approaches that explicitly combine hardware and intelligence. The proposed engineering advantage must be demonstrated in reduced setup, broader supported variation, fewer interventions or better cost, not inferred from the phrase “AI-native.”

Skill execution and recovery



A completed recovery can preserve autonomy; a manual rescue cannot be counted as autonomous success.

Figure 11. Proposed skill-level control. Retries are bounded and unsuccessful attempts remain visible in the production record.

A skill checks its preconditions, executes an action while observing material state and tests the outcome. Recoverable deviations can lead to re-observation, a controlled regrip, recut or rerouting. An uncertain or dangerous state leads to a safe stop. The retry budget should consider fabric damage and elapsed time; unlimited attempts can destroy both quality and shift output.

Research on phase-conditioned imitation learning reports autonomous recovery in T-shirt hanging and removal. This is useful evidence that recovery can be part of a learned controller, but it is not a demonstration of repairing sewn defects or operating this cell. The manufacturing skill set requires its own data, checks and validation. [29]

Training, simulation and deployment

A practical development process collects demonstrations and failures on representative fabrics, parts and intermediate assemblies. Simulation supports geometric checks, policy exploration and scheduling, with a measured gap to real materials. A test set should include withheld parameter combinations, changes in illumination, material lots, edge curl and plausible misalignment. Good performance on repeated training pieces does not establish generalization.

Policies and models remain versioned. Production machines should not autonomously rewrite their qualified behavior after each garment. New versions are tested offline, compared with baselines, released deliberately and reversible. Controlled remote updates, rollback and a local ability to stop safely are product requirements. Model confidence must be calibrated against actual error, especially where a false-positive quality check could pass a defective garment.

Latency and failure independence

Critical motion and sewing synchronization remain local. Network loss should not create uncontrolled movement or leave the plant unable to reach a safe state. A local execution package can continue an already approved job when that behavior is explicitly designed and qualified; otherwise it pauses safely. Cloud services can provide design inference, retrieval, fleet analytics and planning without being the only path to safety.

The general AI risk-management process should document foreseeable failures, data limitations, monitoring and release controls. NIST's AI Risk Management Framework is a useful voluntary governance reference, not machinery certification or proof of safe robot behavior. [26]

13 / Scheduling, capacity and scalable output

A production cell can work on several garments at once. One item may be receiving a collar while another is cut and another is hemmed. This separates the elapsed time for an individual garment from the interval between completed outputs. The proposed architecture should exploit that overlap without exceeding shared transport or buffer capacity.

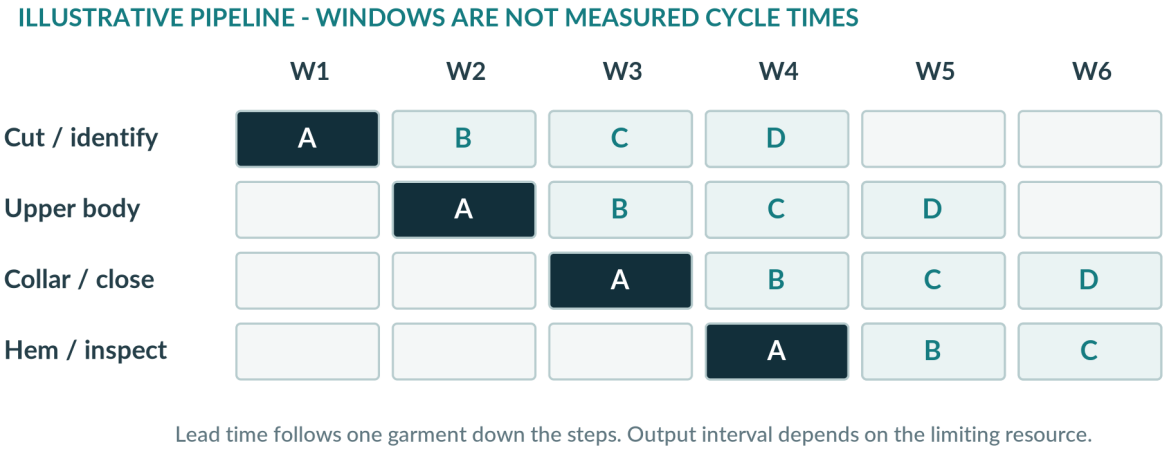


Figure 12. Illustrative pipeline. A–D are different garments; windows are not timed claims. A real model must reserve every station and shared handling resource.

Output arithmetic

For a simplified steady-state model, let H be available shift minutes, A the fraction effectively available for production, Y the acceptable yield and c the running interval between completed attempts. Then acceptable output is approximately **G = H x A x Y / c**. The factors must be defined so a delay or reject is not deducted twice. If c already measures the all-in achieved interval, additional availability or performance deductions may be inappropriate.

Accepted shirts per 12-hour shift	Average accepted interval over the full shift	Required running completion interval at A = 0.80 and Y = 0.95
100	7 min 12 sec	5 min 28 sec
200	3 min 36 sec	2 min 44 sec
300	2 min 24 sec	1 min 49 sec
500	1 min 26 sec	1 min 6 sec
1,000	43 sec	33 sec

These are capacity requirements, not production forecasts. The base 100-shirt objective is a first validation milestone, not a calculated ceiling. The higher scenarios need their own physical configuration, material capacity, cost and reliability evidence. They do not imply the same $100,000 system will deliver every rate.

Resource capacity and bottlenecks

For each resource, calculate total active occupancy per garment, including repeated visits, load/unload and required setup. Divide by the number of genuinely parallel copies. The largest resulting workload is a lower bound on the running completion interval; synchronization, failures and queues can make actual

performance worse. A robot shared across stations can be the bottleneck even when every sewing head is individually fast.

For illustration, suppose a collar module requires five minutes per shirt while all other resources can sustain a two-minute interval. One collar station limits the simplified model to about 109 acceptable shirts per shift; two independently operating collar stations reduce that bottleneck to 2.5 minutes and allow about 219. Reducing the limiting interval to two minutes permits about 274 under the same assumed losses. These are synthetic operation times, not measured collar performance.

JUKI advertises 2,400–3,000 bottom-hemming operations in nine hours with one operator and one machine. This concerns one finishing step with human loading, not complete shirts. It illustrates why local sewing capacity can exceed the intended cell rate while handling and other construction operations remain limiting. [10]

Cold starts, buffers and unfinished work

A full-shift test begins from a declared state: loaded rolls, conditioned or unconditioned inputs, threaded tools, empty or pre-filled buffers and installed software versions. Required in-cell preparation, pipeline filling and end-of-shift work in process affect completed output. A steady-state interval measured after the cell is already full is not the same as a cold-start roll-to-output count.

Buffers absorb variation but consume space and material. In a stable hypothetical pipeline completing an attempt every two minutes with a 20-minute manufacturing lead time, average work in process is approximately ten garments. That relationship is a planning example; actual routing, rejects and multiple visits can require different capacities. An additional buffer does not increase capacity when a single resource remains fully occupied.

Mixed orders and changeover classes

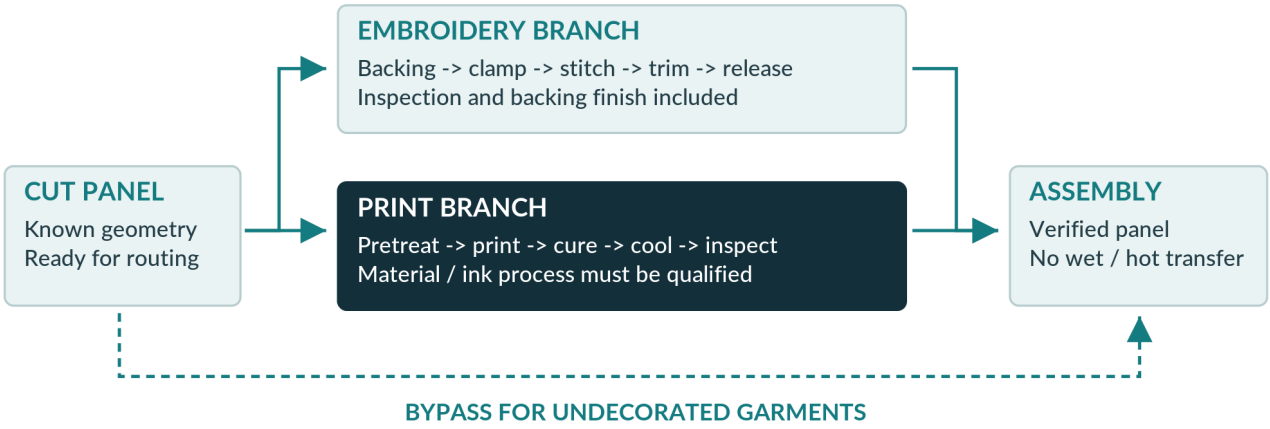
Change class	Proposed treatment
Same material, tools and recipe; new approved size	Recompile parameters and marker; confirm changes remain inside the qualified range.
Same construction, new fabric lot	Verify material profile and any required calibration before release.
New color or trim	Reserve corresponding rolls, threads and supplies; perform disclosed manual or automatic setup.
New construction feature	Check tooling, support access, skill availability, quality and recipe qualification.
Decoration enabled	Reserve complete process capacity, dwell and cooling; use a separate output rating.

The commercial benchmark should compare one order of 100 with ten orders of ten using defined differences. Record preparation, physical changes, blocked time and accepted output. A faster sewing operation does not eliminate expensive style setup. Conversely, maintaining high capacity by suppressing all design variation would defeat the intended small-batch proposition.

14 / Optional printing, embroidery and other processes

Optional modules extend the process without making the undecorated base garment incomplete. Each module needs material compatibility, automatic presentation, supplies, safe handling, quality checks and

a timed interface to the rest of the cell. The production scheduler reserves its capacity before cutting work that depends on it.



A roll-print route would occur before cutting; all branches need independent timing and utility budgets.

Figure 13. Proposed panel-decoration branches. The undecorated path bypasses optional equipment; required preparation and fixing remain inside each branch.

Embroidery

A suitable panel can be embroidered before it becomes difficult to access in a closed garment. The complete process includes design digitization, stitch and density checks, correct backing, automatic clamping or hooping, thread management, stitching, trim operations, backing finishing where necessary and release. The location must be registered to the actual panel and sewing allowances.

Commercial multi-needle embroidery equipment provides a process precedent; Tajima's SAI, for example, has eight needles and specified frames. That does not supply automatic panel loading, stabilization or downstream garment transfer. Those remain integration work, and machine compatibility with thicker stacks or particular backing materials must be tested. [12]

The design must not create an unhandled manual backing-removal step. Thread breaks, color changes, needles and empty supplies also matter to unattended operation. Dense embroidery may dominate cycle time; adding a second head helps only when loading, power, frames and transport can support concurrent work. The base shirt's output rating cannot be applied to arbitrary embroidery.

Direct print, transfers and roll print

Candidate routes include printing a flat panel before sewing, printing qualified roll material before cutting, or using a finished-garment station where loading is feasible. The chosen route depends on artwork, substrate, registration, heat exposure and the desired finish. Ordinary cotton garments cannot simply be assumed compatible with every print chemistry.

Brother's GTX300 documentation describes pretreatment and drying requirements, including pretreatment for white-ink direct-to-garment work. Therefore a proposed print module must include all required treatment, printing, fixing, cooling and inspection for the selected process, not only a printhead. Specific temperature and dwell settings are recipe-dependent and must be validated on the actual material and equipment. [13]

Transfer-based processes add adhesive or powder handling, placement and pressing requirements where used. Roll printing can change the material's dimensions or surface before cutting, requiring subsequent registration and conditioning. Hot, wet or uncured panels must not be sent directly into grippers or sewing stations without a qualified handling state.

Other functions and process ownership

Decorative stitching, selected heat transfers, special closures, retail pressing, folding and packaging are possible modules to study. A garment requiring a closure or interfacing to complete construction must include that operation in its advertised production configuration even if the machinery is physically modular. “Optional” should mean optional to the chosen product, not missing from its promised output.

Each add-on specification declares input state, material range, consumables, heat or air requirements, exhaust, cycle-time distribution, inspection, fault paths and release state. A bypass should maintain base operation when a decoration module is absent or unavailable, provided the current job does not require it. Routing a decorated job around a failed module and calling it complete would be a specification failure.

15 / Quality, reliability and honest autonomy

Quality is defined against an approved sample and technical record. A garment that resembles the intended shape can still fail on seam placement, collar recovery, puckering, skipped stitches, trim location or dimensions. Online inspection and offline qualification have different functions; neither can be replaced by a persuasive image.



Figure 14. Three independent measurement systems: machinery performance, garment quality and customer economics.

Inspection and qualification

The proposed inline checks include identity and lot verification, cut geometry, layer count, seam path, thread presence, visible defects, collar/cuff symmetry, labels and completed construction. Confidence thresholds and known blind spots are documented. Some defects require tactile or force observations; others are not reliably observable from the installed cameras and must be managed through process qualification and sampling.

Offline tests should evaluate dimensions and appearance after a defined wash/dry protocol, seam integrity, stretch recovery, fastening performance and decoration durability as relevant. ISO 6330 provides a reference framework for domestic washing and drying procedures for textile testing; citing its public scope does not establish compliance or select a complete acceptance protocol. A competent textile laboratory and garment specialist should define the actual methods. [25]

Fault classes and accounting

Event	Proposed system response	Performance consequence
Edge shift or uncertain grasp	Re-observe, expose and regrip within a tested limit.	Recovery time is included even when the garment succeeds.
Wrong or damaged cut component	Isolate it; request an identified recut; preserve garment tracking.	Material loss, recut time and blocked downstream capacity.
Sewn defect	Reject or use an explicitly qualified repair route.	Manual repair cannot count as autonomous success.
Thread, needle or mechanical fault	Detect and enter a defined safe state; restart only through a validated recovery.	Any required person counts as an intervention and affects availability.
Low supply or full output	Predict from job requirements and sensed remaining capacity; stop safely if needed.	Refill and clearing time must match the advertised operating model.
Uncertain quality	Hold for designated review rather than silently accept.	Pending review is not an accepted unit at shift end.

A machine may remain unattended by rejecting a problem garment and continuing, but accepted yield falls. A remote operator recovering a job is still human participation. A safety stop is responsible behavior; it simply cannot be hidden when reporting a twelve-hour unattended batch.

Why small failure probabilities accumulate

Under an illustrative independent-trial model, if each garment has a 1% chance of requiring a person, the probability that 100 garments finish without such an intervention is approximately 36.6%. Across 500 garments it falls to 0.66%. At a 0.1% per-garment intervention probability, those batch probabilities are approximately 90.5% and 60.6% respectively.

To achieve a 95% chance of an intervention-free 500-garment batch under that simplified model, the per-garment probability must be about 0.0103% or lower. Actual faults can cluster by material lot, tool wear or environment, so this is not a reliability prediction. It shows why “usually works” is insufficient for an unattended-shift claim.

Test reports should retain every attempt and log cold-start conditions, software and hardware versions, fabric lots, setup, total work time, recovery, operator minutes, rejects and accepted garments. Results on repeated identical cloth should be separated from mixed lots and recipes. A short highlight video or a sequence with failures removed is not a capacity or reliability dataset.

16 / Safety, maintenance, deployment and data protection

The proposed cell contains needles, cutting tools, driven rollers, moving supports, robots and potentially heat, pressurized air, chemicals and stored energy. A machine using a robot marketed as collaborative is not automatically safe as a complete cutting-and-sewing system. Guarding and safe access must reflect the integrated hazards, including foreseeable misuse and maintenance.

ISO 10218-2:2025 provides a public-scope reference for industrial robot applications and cell integration. ISO 12100 provides the general machinery risk-assessment and risk-reduction framework. These references guide engineering review; this white paper is not a compliance assessment, and

applicable standards and legal requirements must be confirmed for the deployment jurisdiction and final equipment configuration. [23,24]

Proposed safeguards and service requirements

Area	Requirement to establish in engineering
Moving and cutting hazards	Guarded zones, controlled access, emergency stop behavior and verified stopping functions.
Stored energy	Safe isolation and release for electrical, pneumatic, vacuum, spring and thermal sources during service.
Fabric entanglement	Detection and safe clearing of wrap, trapped garment, double feed and pulled thread paths.
Hot or chemical processes	Qualified temperatures, ventilation, exposure control, fire protection and compatible handling.
Lint and waste	Accessible collection, cleaning intervals, blockage detection and containment near moving or hot parts.
Roll handling	Maximum load, lifting/support procedure, loading access and secure roll retention.
Remote operation	Permissioned commands, locally enforced limits and no unsafe dependence on network or model availability.
Maintenance	Replaceable wear parts, calibrated replacements, service instructions and recovery testing after work.

Thread paths, blades, needles, gripper surfaces, cameras and filters need inspection and replacement. A cost model that excludes routine servicing is incomplete. Where cleaning or consumable service is required during a shift, the advertised operating model must include it. The machine must be designed so a technician can reach the fault safely rather than dismantle a large portion of the line for a routine repair.

Commissioning and acceptance

A factory acceptance test verifies the supplied configuration before shipment. A site acceptance test confirms installation, utilities, guards, calibration, example recipes and production records under site conditions. Neither is replaced by a supplier video. The delivered system should include the exact capabilities, setup conditions, exclusions, spare-parts list, support expectations and a process for qualifying new materials or tools.

The product should separate user-adjustable parameters from protected machine and safety configuration. Production changes require an audit trail. A restored backup must preserve approved recipes and the physical calibration to which they refer. A replaced camera or gripper can invalidate prior calibration even if the software version is unchanged.

Data, security and customer confidence

Patterns, measurements, reference images and production records can be commercially or personally sensitive. Role-based access, encryption where appropriate, controlled retention and explicit training-data permission are proposed requirements. Project separation matters when one manufacturer serves competing labels. Firmware and policy updates need authenticated delivery, version compatibility and rollback.

Telemetry should help diagnose faults without unnecessarily exposing customer designs. The machine should record its own observations and decisions in a usable form; unexplained model output is a weak service interface. Safety-critical controls remain independent of the conversational system, while general model-risk documentation can follow a governance framework such as NIST AI RMF. [26]

17 / Use cases and deployment models

The most useful customer is not necessarily the person most excited by automatic clothing. It is an operator with identifiable orders, an approved garment specification and a measurable reason to change production methods. Use cases should be evaluated against real alternatives: a capable conventional sewing team, outsourced small-batch production, decorated blanks or a specialist automated service.

Priority use cases to investigate

Use case	Proposed value	Conditions for a convincing deployment
Small contract manufacturer	Pool orders from multiple labels and reduce recurring setup and handling work.	Sufficient aggregate demand, compatible materials, repeatable recipes and service capability.
Established independent label	Replenish proven styles in smaller quantities and adjust sizes or fit without ordering a large finished inventory.	A stable core assortment, known quality standards and volume sufficient to use the equipment.
Uniform or teamwear supplier	Produce supported size variations and recurring replacements, with optional decoration.	Material consistency, approved colors/trims and decoration capacity matched to orders.
Regional production hub	Serve several brands near their customers, coordinating shared raw material and capacity.	Competitive total cost including materials, labor, energy, logistics and any required finishing.
Product-development or sampling center	Convert approved design changes into repeatable physical samples more quickly.	Broad enough tooling and high value from iteration; throughput alone may not justify ownership.
Fashion education or research facility	Explore computational design, production planning and textile robotics.	Appropriate supervision, safe access and a budget justified by learning or research rather than garment margin.

These are hypotheses for customer testing, not estimates of demand. Interviews should collect actual order files, samples, costs, minimums, delivery constraints and reject experience. Interest in a universal clothing machine does not validate the restricted first configuration. A useful customer commitment identifies the required garment, quantity, quality, operating involvement and price conditions that would make adoption rational.

Ownership and production-service models

A manufacturer can buy a cell and sell garments or production services. An operator can own several cells and pool work from brands. An established brand can own equipment for repeat products and outsource unsupported constructions. The technology supplier can sell hardware, software and service without operating garment factories itself. These models have different risks and profit pools; their margins should not be combined as if they accrue to the same business without added costs.

A production-as-a-service offer can reduce the buyer's responsibility for maintenance and utilization, but those costs move to the service operator. A hardware sale can give a manufacturer more control, but it

requires installation, trained staff, spares and a dependable service route. This paper models a hardware supplier and a garment-producing operator separately rather than assuming one model dominates.

Situations where the system may be a poor fit

A business with infrequent orders, a different unsupported construction every day or no verified demand may have low utilization. Very large repetitive orders may favor established dedicated equipment. Products whose economics are dominated by custom fabric development or special washing may gain little from automating sewing alone. Couture-like one-off handling, highly unusual textiles and complex tailored construction are outside any unqualified claim of general capability.

The appropriate comparison is a matched basket of garments and service requirements. A machine making plain knit tops cannot be compared with the price of fully decorated, washed and packed retail products. Similarly, a cheaper outsourced quote may omit shipping, duties, inspection, sampling or inventory commitments. The test must normalize the scope without assuming that local automated production always wins.

18 / Comparable companies and competitive positioning

The opportunity is not based on an absence of innovation. Several companies address automated garment construction, flexible production or design-to-manufacturing software. Their public descriptions differ in joining method, material starting state, autonomy boundary and commercial offer. The following is a research snapshot, not a ranking or independent performance audit.

Company or approach	Publicly described direction	Relevance and boundary of comparison
SoftWear Automation [14]	Machine-vision sewing worklines and on-demand T-shirt production.	Directly relevant to adaptive sewn garments. The reviewed offer emphasizes sourcing through enabled factories; it does not establish this paper's turnkey price and full configuration.
Silana [15]	Cutting, robotic layer separation and adaptive autonomous sewing without templates or fixtures.	A close mechanical competitor. Its page also states one worker per ten machines; advertised automation should not be treated as verified zero-intervention lifetime operation.
CreateMe [16,17]	Robotic assembly centered on digital adhesives, specialized tooling and Physical AI.	Explicitly co-designs mechanics and intelligence. Bonding differs from the proposed conventional stitching. Its research page labels Gen 1 in production and Gen 2 in development; its brands page advertises minimum order/color quantities of 100.
Resonance / CreateOne [18]	Design intent compiled into manufacturing instructions, with production-network orchestration and feedback.	Closely overlaps the digital design-to-production thesis. Public claims do not establish an equivalent standalone fully autonomous cut-and-sew cell.

Company or approach	Publicly described direction	Relevance and boundary of comparison
Sewbo [19]	Temporarily stiffens fabric so industrial robots can handle and sew it, followed by removal of the treatment.	A meaningful process redesign, but it adds a material-treatment route outside the proposed untreated-textile baseline. Its site says development and partner trials continue and the technology is not yet available for purchase.
SHIMA SEIKI WHOLEGARMENT [20]	Whole-garment knitting directly from yarn.	An important digital garment-manufacturing alternative, but not a substitute for every conventional woven or jersey cut-and-sewn product.
Printful [21]	No-minimum customization within selected products, including all-over-print cut-and-sew offerings.	Demonstrates an accessible low-minimum service model. A product catalogue is not unrestricted customer control of patterns, fabric and assembly.
Process suppliers [01,02,10–13]	Cutting, component sorting, material preparation, sewing, printing and embroidery equipment.	Potential suppliers, integration partners or adjacent alternatives. Individual functions do not demonstrate the complete proposed cell.

Why “built for AI” is not a sufficient distinction

CreateMe’s research description is particularly relevant: it links single-sided access, tooling, grippers and bonding to learning-based control. That directly contradicts a claim that competitors have only added AI superficially to legacy machines. Its stated roadmap must still be distinguished from demonstrated broad capability. [16]

The same discipline applies to every company. Public marketing language is not proof of rated speed, material breadth or cost; a company’s silence about a feature is not proof it lacks it. No source reviewed here establishes an independently verified equivalent matching all of this paper’s requirements at approximately $100,000. That is a limit of the reviewed public evidence, not a claim that no private development exists.

The proposed differentiation to establish

Proposed distinction	Evidence needed to make it credible
Complete roll-to-output continuity	A run including necessary preparation, cutting, component extraction, assembly, finishing and discharge without hidden handling.
Conventional stitched construction	Garments matching specified seams, ribbing, fabrics and appearance rather than substituting another joining process without approval.
Reusable skills across garments	Materially different supported designs on the same core platform, with documented setup, tooling and qualification changes.
Low small-order overhead	Measured job preparation and changeovers across a mixed-order shift, not only fast identical-item production.
Deployable product for smaller operators	A costed delivered configuration, accessible service, transparent utilities and installation requirements.
End-to-end digital consistency	Design edits correctly update patterns, technical records and production jobs; physical observations remain linked to the approved revision.

These distinctions form a **comparison agenda**, not established competitive superiority. The strongest demonstration would benchmark a representative order mix against both conventional small-batch production and relevant automated alternatives using the same garments, material quality, delivery scope and human involvement.

19 / Implications for minimum orders and the industry

The most important economic effect could be reducing the cost of beginning another small job. Faster stitch formation alone is not sufficient. Pattern approval, material procurement, machine setup, labor allocation, decoration and order administration all contribute to small-order cost. Some are addressable by the proposed system; others remain upstream or outside its process boundary.

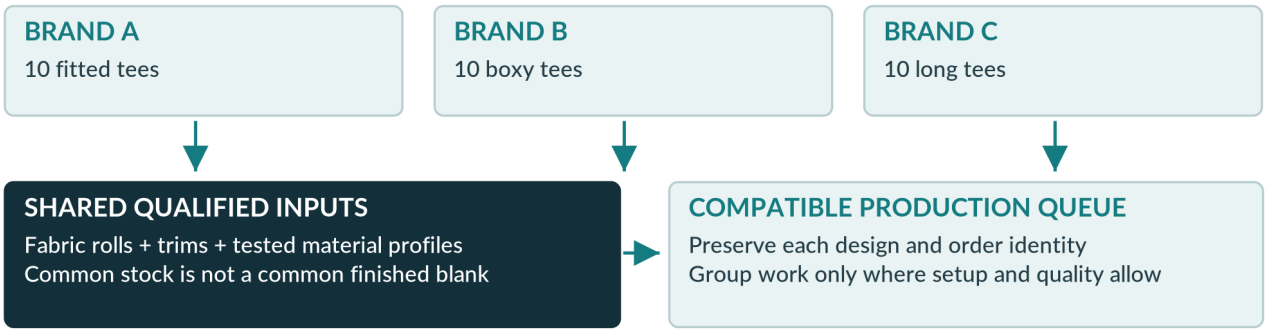


Figure 15. Proposed pooled-production model. Small orders can share qualified raw materials and capacity without sharing an identical finished blank.

Which minimums can change

A software-qualified design variation can potentially reuse a construction recipe and reduce setup. Automated cutting and identified transfer can reduce manual kit preparation. Compatible jobs can share a marker and be scheduled around common materials and tools. These are proposed routes to lower per-order work, not proof that every new style needs no preparation.

Fabric and trim minimums do not disappear. Pine Crest Fabrics, for example, distinguishes a 15-yard minimum for stocked fabrics from typically 300–500 yards for made-to-order materials. This illustrates the difference between drawing from shared stock and commissioning a dedicated upstream run. The proposed machine changes garment production, not the mill's process. [22]

Constraint	Possible effect of the proposed system	Residual issue
New pattern or fit	Parametric tools and reusable approved recipes can reduce repeated drafting work.	Fit and construction still require qualification; photographs do not establish exact measurements.
Sewing setup	Configurable supports, skills and tooling can reduce style-specific preparation.	Unsupported seams and materials may need new physical development.
Cutting and kit preparation	Integrated layout and extraction reduce manual handoffs.	Material preparation, scrap and pickup reliability remain real work.
Fabric color or development	Shared stock can pool demand from several customers.	Custom material runs and shade requirements retain upstream constraints.

Constraint	Possible effect of the proposed system	Residual issue
Decoration and special finishes	Qualified modules can reduce coordination and some handling.	Their own setup, dwell, consumables and material minimums remain.
Order management	Consistent jobs and traceability can reduce administrative repetition.	Customer approval, exceptions, payment and quality disputes still have a cost.

The setup-cost mechanism

As a simplified illustration, $200 of style-preparation cost adds $10 per item to a 20-piece batch, but only $1 to a 200-piece batch. Reducing that preparation to $20 reduces those additions to $1 and $0.10 respectively. The value comes from lowering setup, not merely distributing the same setup over more units. These values are not industry benchmarks.

The cell's scheduling rules should therefore aim to minimize the expensive changes that actually exist: rolls, thread colors, tools, cure conditions or unqualified construction details. Jobs can be grouped when compatible without changing the customer's approved garment.

Inventory and working stock

Smaller finished-garment runs can shift some stock from style-specific finished goods toward reusable fabric and trims. That does not eliminate inventory risk; the operator or supplier now carries more shared material and can still face obsolete colors, excess widths or unused specialty trims. The economic test includes remaining roll value, unused cut pieces, minimum purchasing quantities and carrying costs where material.

Smaller replenishment may reduce unsold finished stock or cash tied to large orders. Measure those customer-specific effects from actual demand; do not add assumed retail margin to the manufacturing benefit.

Labor, skills and production geography

The proposed system changes the allocation of work rather than establishing a labor-free factory. Setup, garment engineering, material qualification, quality review, maintenance, logistics and customer operations remain. The operator model in this paper deliberately includes paid work. Whether automated assembly reduces total labor cost depends on how much service and exception handling it introduces.

Local or regional production becomes attractive only when total economics support it. Energy, rent, service, material supply, utilization and logistics still matter. A modular cell may enable a different production footprint, but it does not guarantee lower landed cost in every country or displace all large-scale conventional manufacturing.

Environmental effects require measurement

Potential benefits include lower finished-goods overproduction, fewer unnecessary transfers and better traceability. Potential costs include energy for robotics, vacuum, air and heat, electronic equipment, reject fabric, additional chemistry and more frequent small shipments. Easier production can also stimulate more consumption. No percentage reduction in emissions or waste is assumed.

An evaluation should compare the full lifecycle and product durability, not only factory electricity or one shipping leg. UNEP's textile value-chain roadmap treats sustainability as a chain-wide issue; automation is one possible contributor rather than a complete solution. Measure material utilization, rejected mass, energy per acceptable garment, consumables, shipment patterns, repairability and garment life under a consistent boundary. [27]

20 / Machine cost, selling price and supplier margin

The approximately **$100,000 selling price** is a design objective for a defined complete base configuration, not an engineering quotation. Cutting, extraction, construction, required controls and guarding must be included. Optional decoration, unusual site work and additional high-capacity modules need separate stated prices. A lower price achieved by quietly excluding automatic collar preparation or final construction would not meet the product definition.

What these estimates mean

The following is an **illustrative should-cost envelope** prepared to expose cost pressure. It is not a bill of materials, a set of current vendor quotes, a probability range or proof that the lowest figures are achievable. All three columns intend the same core end-to-end scope; they represent different procurement and integration outcomes, not different levels of advertised autonomy.

The optimized column assumes substantial integration, repeatable assembly and carefully selected custom mechanisms. The reference column allows more complete bought-in subsystems. The conservative column represents more expensive equipment and integration. None presumes a specified production volume until suppliers validate it. Costs must eventually be quoted at explicit build quantities and include documented interfaces and service obligations.

Base-system cost category	Optimized repeat-build objective	Reference integration case	Conservative procurement case
Roll handling and material preparation	$4,000	$7,000	$12,000
Single-ply cutting and hold-down	$8,000	$14,000	$25,000
Part extraction, grippers and buffer	$5,000	$8,000	$14,000
Manipulation and shared transport	$8,000	$14,000	$25,000
Sewing, trim and common forming/closure tools	$10,000	$15,000	$25,000
Vision, sensors, control and compute	$4,000	$6,000	$9,000
Frame, guarding and base utilities	$6,000	$9,000	$15,000
Assembly, calibration, acceptance test and packaging	$7,000	$10,000	$15,000
Factory build-and-test cost	**$52,000**	**$83,000**	**$140,000**

The tooling allowance is particularly uncertain. A configuration claiming specific blouse closures, rib cuffs or construction fusing must contain those functions. If selected tools exceed the allowance, the cost model changes; required operations cannot be reclassified as manual work to preserve the price. Long-duration material conditioning or autonomous roll/thread changes may also demand more equipment than these allocations support.

A low advertised robot price is not an adequate basis for estimating the cell. UFACTORY's Lite 6 is advertised from $2,999, but its published payload is 600 g and reach is 440 mm. Its suitability depends on tooling mass, cloth load, reach and dynamics. The price is a useful example of why a component headline must be read with its limits, not a component selection for this design. [28]

Contribution at a $100,000 system price

Assume factory build costs as above, $6,000 per unit for delivery and commissioning, a warranty provision equal to 5% of selling price and variable selling/channel costs of 4%. These are model assumptions. They are not validated freight, installation, warranty or dealer terms.

Per delivered system	Optimized case	Reference case	Conservative case
Selling price	$100,000	$100,000	$100,000
Factory build and test	($52,000)	($83,000)	($140,000)
Delivery and commissioning	($6,000)	($6,000)	($6,000)
Warranty provision	($5,000)	($5,000)	($5,000)
Variable selling/channel allowance	($4,000)	($4,000)	($4,000)
Contribution before fixed company costs	**$33,000**	**$2,000**	**($55,000)**
Contribution as a share of revenue	**33%**	**2%**	**–55%**

The contribution must still cover sustaining engineering, software operations, fixed service capacity, administration and other company expenses. Original development expense, taxes and any financing costs are not recovered by merely labeling contribution “profit.” Early field failures, rework, returns or unplanned installation can make the allowance inadequate.

Price sensitivity and a cost ceiling

With these assumptions, contribution is **0.91 x selling price – factory cost – $6,000**. To obtain a 30% contribution rate, the required selling price is approximately **(factory cost + $6,000) / 0.61**.

Factory-cost scenario	Price for 30% contribution under this model
$52,000 factory cost	$95,082
$83,000 factory cost	$145,902
$140,000 factory cost	$239,344

For the reference $83,000 factory cost, a $125,000 selling price produces $24,750 contribution, or 19.8%. A $150,000 selling price produces $47,500, or 31.7%. Those numbers are arithmetic, not proposed market prices or evidence of willingness to pay. They show that the $100,000 target requires a credible path to a significantly lower factory cost than the reference case.

At a $100,000 selling price, the model permits a factory cost of roughly $55,000 if a 30% contribution rate is required. That should become a subsystem allocation and supplier-quotation exercise. Reducing hardware spend is not a success if it causes more service calls, lower accepted output or hidden manual steps.

Repeatability, recurring service and company profitability

The manufacturer’s economics depend on repeatable installation rather than a new bespoke engineering project for each site. Common hardware, controlled configurations, calibration procedures, spare parts and remote diagnostics can reduce cost variability; those advantages must be demonstrated across installations.

For an illustrative annual scale check, 50 systems at $33,000 contribution generate $1.65 million before fixed company expenses. At $2,000 contribution they generate only $100,000. If annual fixed operating

expense were $1.5 million, those scenarios would leave $150,000 and negative $1.4 million respectively before omitted development recovery and other excluded items. The sales volume and overhead are assumptions, not a forecast or a development plan.

A separately priced software/service package could contribute revenue, but it also incurs compute, updates, service staff and support costs. For illustration, a $3,000 annual package costing $1,800 to deliver contributes $1,200 before fixed costs. Warranty coverage and paid service must be defined so the same work is not charged or costed twice. Recurring fees should not be used to conceal an unserviceable machine or assume that every buyer will pay indefinitely.

21 / Garment cost: a transparent premium T-shirt model

This section models the **operator who produces garments**, not the equipment supplier. It uses a fully disclosed premium blank T-shirt example. It is intentionally not based on a $60–$100 retail selling price: a factory's manufacturing value and a brand's retail margin are different. Actual material quotes, staffing, utilities and customer orders must replace the assumptions before commercial decisions.

Model boundary

The garment is an undecorated heavyweight cotton T-shirt with ribbing, normal labels/reinforcement and approved conventional construction. Body and rib material are purchased; their cutting waste and rejected attempts are included. The assumed ex-factory selling price is $12 per acceptable garment. Taxes, debt service, customer delivery, brand marketing and specialty finishing are outside this example. Basic packaging is included even though it is outside the proposed automated construction boundary.

The model assumes 250 operating days per year. Each day includes the stated paid work and overhead whether 25 or 100 shirts sell. Output variables below mean **acceptable garments both produced and sold**, not merely machine capacity. The quality loss is already inside variable cost and must not be deducted again from those accepted-unit counts.

Input assumptions and material mass balance

Input	Assumption
Net body-panel cut area, including seam allowances	1.15 m^2 per shirt
Body fabric mass per area	0.280 kg/m^2, equivalent to 280 gsm
Net rib cut area	0.075 m^2 per shirt
Rib mass per area	0.360 kg/m^2
Area utilization of each cutting marker	85%
Body and rib purchase prices	$12/kg and $14/kg respectively; unquoted scenario values
Trim allowance per attempted garment	$0.45 for labels, reinforcement and related specified trims
Thread and routine small consumables	$0.15 per attempted garment
Final acceptable yield	95%; rejected garments receive no salvage credit
Basic packaging	$0.20 per acceptable garment

Body material required per attempt is **1.15 x 0.280 / 0.85 = 0.3788 kg**. Rib material required is **0.075 x 0.360 / 0.85 = 0.0318 kg**. Their combined purchase cost is **$4.9906 per attempt**. Adding trims and routine consumables gives **$5.5906**. Dividing by the 95% acceptable yield and adding packaging produces **$6.0848 per acceptable garment**, rounded to $6.08.

This is a conservative full-attempt reject model with no salvage credit. In practice, an early detected cut defect can cost less than a fully rejected garment, while a late defect, partial recut or manual rework can have a different cost. A production model should replace this simplification with event-level material and time records.

The modeled input body-plus-rib fabric per acceptable shirt is approximately **0.4322 kg**. At 100 accepted shirts, that is **43.22 kg** including marker loss and quality rejects. It is not the weight of the finished garment. This distinction matters for material supply, roll capacity, scrap handling and cost claims.

Daily operating costs

Expense	Basis	Daily allowance
Paid work	6 hours at $20/hour, including setup, replenishment, inspection, packing and allocated order handling	$120.00
Electricity	5 kW average across 12 hours at $0.15/kWh	$9.00
Space, insurance and basic administration	Allocated daily cost	$70.00
Maintenance reserve	$5,000 per year across 250 operating days	$20.00
Software/service	$3,000 per year across 250 operating days	$12.00
Daily fixed cash operating cost		**$231.00**
Depreciation	$110,000 equipment plus site basis, seven-year life, zero residual, 250 days/year	$62.86
Daily fixed cost including depreciation		**$293.86**

The $110,000 depreciation basis comprises an assumed $100,000 delivered machine plus $10,000 of site preparation. This is an accounting illustration, not tax advice or a validated installation quote. The supplier's delivery allowance and the operator's site work are different items. The model excludes a financing arrangement and any location-specific tax or duty treatment.

Six paid hours do not mean a person sews for six hours inside an autonomous run. They recognize that loading, inspection, packaging, order management and service availability still require paid effort. The exact staffing arrangement must provide safe access and adequate response; higher labor is tested in Section 22. The 5 kW average is an energy assumption, not a measured system specification, and excludes thermal decoration modules.

Cost and operating profit at different sales volumes

Operating cost per acceptable shirt is **$6.0848 + $293.8571 / daily accepted sales**. Daily operating profit at the assumed $12 ex-factory price is **accepted sales x ($12 − $6.0848) − $293.8571**.

Accepted shirts sold per day	Modeled total cost per shirt	Operating profit per day	Operating profit per 250-day year
25	$17.84	($145.98)	($36,494)
50	$11.96	$1.90	$475
75	$10.00	$149.78	$37,445
100	$9.02	$297.66	$74,415
150	$8.04	$593.42	$148,354
200	$7.55	$889.18	$222,294

This table isolates the effect of volume while holding equipment, daily staffing and other assumptions fixed. The 150- and 200-shirt rows are **sensitivity cases, not evidence that the same cell and labor achieve those outputs**. If extra modules, labor, hours, maintenance or power are needed, those rows must be re-costed. Unsold output does not create the revenue shown here.

At 100 accepted sales per day, daily cash operating surplus before depreciation is approximately $360.52, while modeled operating profit after depreciation is $297.66. Neither is after-tax company profit. A crude purchase-cost payback based only on the modeled surplus would be misleading without considering demand, ramp-up, working stock, service reserves and the buyer's actual alternative. Those items should be modeled explicitly for a real deployment.

22 / Sensitivities, profit boundaries and decision tests

The operator model's most important lesson is the dependence on **accepted, sold volume and total cost**, not a particular profit number. Several plausible changes can erase or materially reduce the modeled margin. These calculations hold all other assumptions fixed and therefore show sensitivities, not independent forecasts.

Selling price and utilization

At 100 accepted shirts sold per day, the modeled total cost remains $9.02 per shirt. The assumed ex-factory selling price changes the result as follows.

Ex-factory price per accepted shirt	Operating profit per day	Operating profit per year
$8	($102.34)	($25,585)
$10	$97.66	$24,415
$12	$297.66	$74,415

At the $12 selling-price assumption, break-even is approximately **49.7 accepted sales per day**. At $10 it is approximately **75.1**. A machine capable of producing 200 units but selling only 50 should be evaluated at 50 sales with any excess inventory costs, not at nameplate capacity. Conversely, demand greater than capacity does not create profit until the system can supply it reliably.

Material, yield and labor sensitivity

One change from the baseline	Total cost per shirt at 100 accepted sales/day	Interpretation
Body fabric $10/kg instead of $12/kg	$8.23	Requires an actual equivalent-quality material quote.
Body fabric $16/kg	$10.62	Material cost can dominate the automation benefit.
Marker utilization 75% instead of 85%	$9.72	Includes increased body and rib input, not a second quality deduction.
Marker utilization 90%	$8.73	Must preserve grain, matching and defect exclusions.
Acceptable yield 85% instead of 95%	$9.72	Full-attempt reject model; actual fault location changes the cost.
Acceptable yield 98%	$8.84	Must be measured, not assumed from visual examples.
Paid work 12 hours instead of 6 hours/day	$10.22	Adds $120/day at the same hourly assumption.
Delivered machine $150,000 plus $10,000 site work	$9.31	Higher depreciation only; any changed capacity or service cost is not credited.

Real changes can be correlated. A more difficult fabric can reduce marker utilization, slow handling and increase rejects together. A higher output configuration can require more labor and electricity. Combining only favorable independent sensitivities would overstate profit. A mature model should use measured cycle-time and defect distributions by material and recipe.

Decoration and branded-product economics

Decoration contributes a separate selling price and separate costs: ink or thread, backing, pretreatment, transfer media, heat, dwell, module depreciation, rejects and handling. A $5 decoration surcharge is not $5 of profit. If the decoration station constrains output, it also changes the fixed-cost allocation of the entire cell. A higher-value garment can justify slower production, but only when buyers actually pay the corresponding price.

A brand selling a garment at retail must still pay for customer acquisition, transactions, fulfillment, returns, customer service, design and inventory. The operator's modeled $12 ex-factory revenue is not automatically the brand's manufacturing cost in every arrangement, and the brand's retail price cannot be used as a proxy for factory savings. Profit should be attributed to the business that performs the work and carries the risk.

Benchmarking the alternative

The comparison with outsourced production should match fabric, construction, quantity, quality, lead time, decoration, packaging and delivery terms. The comparison with an internal conventional team should include its actual productivity, overhead and flexibility. New-machine benefits may include lower minimums, faster repeat orders, traceability or less finished inventory, but these should be quantified separately rather than used to excuse a weak manufacturing-cost calculation.

A deployment decision should use at least three scenarios: low demand and elevated fault cost; realistic recurring orders; and higher utilization with its required staffing and equipment. The evidence should show what happens when the system is unavailable for several days, a material lot fails qualification or a

customer changes a specification after cutting. Reserves and service coverage should reflect those events.

Two different economic conclusions

Under the reference supplier cost, the $100,000 selling price leaves little contribution even when the operator model appears profitable. Under the optimized supplier cost, the machine business can have room to cover fixed costs, but the operator still needs sufficient accepted sales and a competitive total garment cost. Both models must work; one cannot be used as proof of the other.

The appropriate next costing artifact is a supplier-backed bill of materials and installation scope, accompanied by a production-cost model populated with measured garment trials and real customer orders. The numerical examples here make those questions explicit. They do not establish a profitable product in advance of that evidence.

23 / Validation and product-development sequence

The technical program should progress from the hardest unsupported transitions to complete garments, then to repeatable mixed-order production. It should not begin by optimizing a visually attractive enclosure or treating successful straight seams as proof of a full system. Several construction families must influence the architecture before hardware choices make breadth prohibitively difficult.

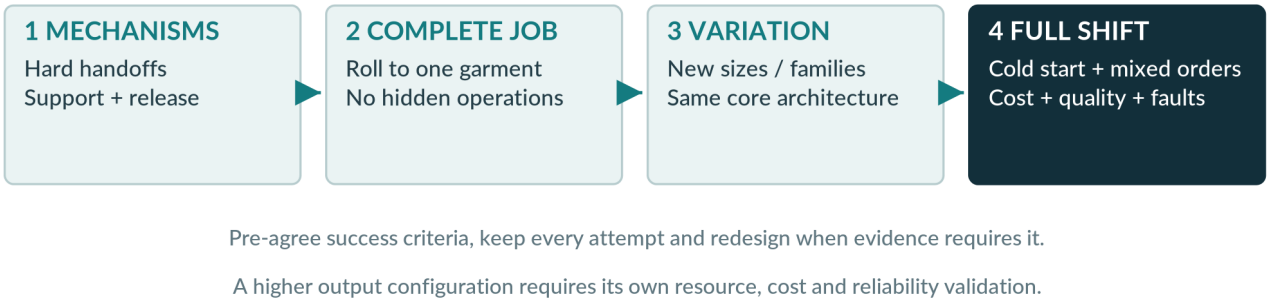


Figure 16. Evidence-led development sequence. Each stage has measurable outputs; higher speed and broader materials require separate qualification.

Stage 1: close the physical gaps

Produce a full support-and-access map for the reference shirt and representative woven and bulky-rib operations. Build real-fabric rigs for extraction, an uncertain edge, a partly assembled garment, a collar or cuff opening, and a turning or closure task. Demonstrate how the source releases and the destination retains control. Compare mechanisms on realistic materials rather than selecting one from a polished drawing.

The output is a chosen architecture with reasons, alternatives and identified limits. Necessary design files include kinematics, tool interfaces, support geometry, sensing, safe-stop behavior and a tentative cost allocation. A manual demonstration can help establish reach and access, but any hand movement not provided by the proposed mechanisms remains an unresolved automation requirement.

Stage 2: connect an uninterrupted complete route

Integrate a defined garment recipe from loaded rolls through accepted or rejected output. Preserve piece identities and material lots across cutting and sewing. Include rib-loop preparation, labels, reinforcement, actual hems, turning and required final construction. Log every operator action. The software should generate the same geometry and recipe used by the cell, not a separate demonstration design.

One successful garment establishes an important integration result but not production reliability. The cell may be slow and use costly prototype equipment at this stage; the report should identify which components need redesign for the intended delivered product and why replacing them is technically credible.

Stage 3: demonstrate repeatability and variation

Agree a test matrix before trials. Include repeated runs, size variations, changed material lots and selected new constructions. Retain failures, reject garments and recovery logs. Assess output after release and after the agreed offline wash and durability tests. A sample-specific adjustment performed between every run must be disclosed; it is not evidence of a general recipe.

A cross-family test should use the same core transport and support architecture on meaningful knit, woven and bulky-rib tasks. Document which changes were parameters, tools, material profiles or new mechanisms. This makes “programmable” a measurable property rather than a marketing label.

Stage 4: measure complete shifts and economic performance

Run repeated 12-hour trials from declared starting conditions. Test both a repeated garment and a mixed-order workload. Record accepted output, startup, pipeline draining, roll preparation, changeover, consumed supplies, energy, paid operator time, automatic recoveries, manual interventions and quality holds. Validate installation and service procedures on more than one physical build before claiming repeatable deployment.

Compare measured costs with the supplier and operator models. Revise price, configuration, output rating or product scope when evidence requires it. Do not retain an attractive target by excluding inconvenient operations or changing the definition of an acceptable garment.

Required technical records

The final engineering record should include mechanical CAD, wiring, module interface descriptions, safety analysis, calibration and maintenance procedures, recipe and policy versioning, a supplier-backed costed parts list, process qualification, full trial data, software tests and known limitations. Customer-facing documentation should state input conditions, setup, supported products, capacity basis, utilities, service and exclusions in plain language.

24 / Conclusions and unresolved questions

The research and existing process equipment support a credible route toward an integrated, software-directed garment-production cell. The proposed product does not require a new textile or a new principle of stitch formation. It does require a substantial advance in connecting cutting, material control, garment topology, sewing access and recoverable execution in a repeatable system.

The industrial format provides useful physical freedom: independent sewing processes, accessible supports, parallel work and modular expansion. The AI contribution is meaningful when it reduces uncertainty and setup effort, preserves design-to-production consistency or recovers from material

variation. AI does not remove physical constraints, replace all deterministic control or establish unlimited designs.

Several relevant companies are pursuing parts of the same ambition, and some explicitly co-design learning and mechanics. The differentiating case therefore rests on measured capabilities: conventional stitch quality, end-to-end scope, useful garment breadth, transparent autonomy, low small-order overhead, cost and serviceability. The paper does not establish a first-in-category claim.

The $100,000 selling-price objective is demanding. Under the illustrative reference supplier cost it leaves almost no contribution, whereas an optimized build could leave materially more room. Garment-operator profitability is similarly sensitive to utilization, material cost, yield and labor. The economic models demonstrate what would need to be true, not what is already true.

Questions that must be answered by evidence

Open question	Decisive evidence
Can the entire garment remain controlled and accessible?	Complete real-fabric sequence including closed openings, regripping and release.
Can the premium finish be preserved?	Accepted garments matching the technical pack, including wash and durability qualification.
Does the same architecture generalize?	Meaningful design and construction variations without hidden redesign or manual rescue.
Can the cell run productively for a full shift?	Repeated declared-start trials with all faults, interventions, consumables and rejects recorded.
Are low minimum orders economically useful?	Mixed-order preparation and changeover results, compared with actual customer alternatives.
Can a repeat-build product support the target price?	Supplier quotes, production assembly data, installation cost and field-service evidence.
Will customers use enough capacity?	Real recurring orders and accepted quality/price conditions, not interest in an unrestricted ideal machine.

> The project is best understood as a proposed manufacturing platform whose value comes from joining known processes with new, verified transitions. Its strongest outcome would be economical, high-quality production of many supported designs in small quantities—not an unbounded promise that any image becomes any garment automatically.

Appendix A / Requirements and verification matrix

The following requirements translate the concept into evidence that can be checked. They are proposed acceptance categories; quantitative limits and sample counts must be established with the selected design and product specification before formal trials.

ID	Requirement	Verification method
R01	Roll-to-output baseline includes cutting and component preparation.	Observe complete runs from declared loaded state; reconcile every material and human handoff.
R02	No hidden pre-assembly or final construction.	Audit incoming materials, recipe and output; inspect collars, hems, labels and closures.
R03	Design, patterns and machine job remain consistent.	Change representative parameters and verify all dependent artifacts and physical outputs.
R04	Material qualification is explicit.	Lot/profile checks and trials at declared operating limits.
R05	Each component retains identity.	Trace random panels through extraction, buffer, recut, sewing and output.
R06	Handoffs preserve support and access.	Instrument transfer acknowledgment; test misalignment and uncertain-grip scenarios.
R07	Garment release is mechanically possible.	Full sequence with closed openings and support removal; no manual extraction.
R08	Quality matches the approved specification.	Inline checks plus offline dimensions, appearance, seam and wash testing.
R09	Recovery is bounded and observable.	Fault injection with time, damage, retry and safe-stop records.
R10	Full-shift output counts acceptable finished garments.	Timed cold-start and steady-state reports with rejects and work in process separated.
R11	Mixed-order setup is transparent.	Compare repeated and mixed jobs, recording tool, material, thread and approval changes.
R12	Multiple garment families inform the platform.	Cross-family tests and documented hardware/software changes.
R13	Optional modules complete their whole process.	Decoration trial from untreated input to qualified released panel or garment.
R14	Safety is independent of general AI and connectivity.	Risk review and validated safe-state testing by qualified personnel.
R15	Service and installation are repeatable.	Acceptance at multiple builds/sites; measure repair access, calibration and downtime.
R16	Price and profit assumptions are replaced by evidence.	Costed supplier parts, installation, support and actual operator production records.

Suggested trial record

Each attempt should record job ID, garment/recipe revision, material lot/profile, tool versions, timestamps, station occupancy, observations relevant to failure, retry events, interventions, material consumption, reject reason and final inspection. Each shift adds setup time, starting buffer state, roll readiness, energy, paid work and final unfinished inventory. Record whether any off-cell preparation occurred and who performed it.

No fixed number of successful demonstrations is treated as sufficient for every claim. The required sample size depends on the failure-rate confidence, material diversity, consequence of failure and intended operating claim. Rare unattended failures need much larger evidence sets than one hundred convenient cycles.

Appendix B / Example machine-readable garment record

This schema is an illustrative information contract, not executable machine control. Identifiers, measurements and recipe names are examples; an implementation would require formal types, unit handling, geometry storage, validation and authenticated authorization.

Field group	Example content and validation
garment_id / revision	TEE-DEMO-001 / 7; immutable once a job is approved.
design_family	crewneck_knit_top; must map to an available construction family.
panels	front, back, left_sleeve, right_sleeve, neck_rib; each with true boundary, seam line, allowance and orientation.
material_assignments	BODY-LOT-12 with approved jersey profile; RIB-LOT-04 with approved rib profile.
seam_edges	Explicit edge pairs, directions, seam type and local correspondence/notches.
rib_relationship	Validated relaxed rib length, body neckline length and permitted local stretch distribution.
trims	Tape, labels and other parts with supply identity, placement and operation dependency.
process_requirements	Overlock, coverstitch, approved rib tool, thread colors and size limits.
assembly_graph	Ordered dependencies with alternative routes allowed only inside the qualified recipe.
entry_contracts	Required supports, exposed seam, observed layer count, ready receiving station and material state.
quality_profile	Sample reference, dimensions/tolerances, seam limits and relevant offline qualification.
job_authorization	Approved revision, cell configuration, allowed quantity, permitted policy versions and operator approval.
results	Accepted/rejected outputs, observations, faults, consumption and actual setup/intervention record.

Example skill contract: transfer a joined upper body

The source holds a garment with completed shoulder seams. The target requires the neckline exposed on an opening support with no unrelated body layer in the sewing path. The skill verifies identity, checks source support, estimates pose, exposes the neckline, positions the receiving support, confirms grip/layers, transfers control and only then releases the source.

Permitted recovery might re-observe, lift a known region or regrip within a limited tension envelope. A twisted or unidentifiable garment goes to a reject/hold route or safe stop. Successful motion without the target condition is not success. The result updates the garment state so the collar operation receives a verified input rather than an assumption.

Appendix C / Equations and financial definitions

Quantity	Definition used in this paper
Simplified accepted shift output	G = H x A x Y / c, with losses defined to avoid double counting.
Running interval required for a target	c = H x A x Y / G.
Shared-resource lower bound	Per-garment active workload divided by parallel copies, including every required visit; take the largest resource value.
Mean work in process in a stable flow	WIP = throughput rate x mean manufacturing lead time, using consistent attempt/accepted units.
Independent no-intervention batch probability	B = (1 − p)^N; actual correlations require a different model.
Raw material per attempt	Net cut area x material mass/area divided by marker utilization, summed by material.
Variable cost per acceptable garment	Attempt-related material/consumables divided by acceptable yield, plus accepted-unit expenses.
Operator cost per accepted sale	v + F/n, where F includes the declared daily fixed expense and depreciation.
Operator operating profit	n x (selling price − v) − F.
Supplier contribution	Selling price − factory cost − delivery − warranty provision − variable selling expense.
Modelled supplier contribution here	0.91 x selling price − factory cost − $6,000.
Price for 30% supplier contribution here	(Factory cost + $6,000) / 0.61.

Factory cost is the modeled cost to build, test and package the equipment. **Supplier contribution** deducts the specified per-sale costs but not fixed company overhead. **Operating profit in the operator model** deducts the declared daily costs and depreciation but excludes taxes, financing and any unmodeled business expense. **Cash operating surplus** adds back depreciation only; it does not account for changes in working stock or equipment replacement.

Yield is defined at the relevant stage. Marker utilization is an area-allocation measure; acceptable-garment yield is an output-quality measure. They are distinct losses and are both included once in the

material model. Where a measured rate already includes the loss, applying the same factor again would understate output or overstate cost.

Money is rounded for presentation; calculations use unrounded values. Annual operator results use 250 operating days. No exchange-rate assumption is required because the examples are entirely in USD. Tax, import treatment, local labor rules and jurisdiction-specific product requirements require separate local analysis.

Appendix D / Assumption register and information still required

Model or design assumption	Current status	Evidence needed to replace it
$100,000 base delivered price	Design objective	Delivered-scope definition, supplier costing and customer comparison.
$52k / $83k / $140k factory cases	Illustrative allocations	BOM, build quantity, vendor quotes, integration labor, yield and acceptance cost.
Delivery $6k; warranty 5%; selling 4%	Scenario allowances	Route/site quotes, service experience, channel terms and return exposure.
100 accepted shirts per 12 hours	Validation objective	Complete-shift tests from disclosed starting conditions.
200–300 and 500 higher-output cases	Configuration studies	Station timings, parallel hardware, additional cost and full-shift results.
80% effective running and 95% quality yield	Capacity-sizing assumptions	Consistently defined measured availability, performance and yield.
1.15 m^2 body; 0.075 m^2 rib	Example pattern areas	Actual approved cutting geometry and marker outputs.
280 / 360 gsm; 85% marker use	Example material and nesting inputs	Material tests, usable width, grain, defects and real layouts.
$12/kg body and $14/kg rib	Unquoted purchase assumptions	Equivalent-quality quotes at intended purchasing quantities and terms.
Six paid hours/day at $20/hour	Operating scenario	Work measurement, coverage needs, exception frequency and local loaded labor cost.
5 kW average; $0.15/kWh	Energy scenario	Metered duty cycle including air/vacuum; actual tariffs and added process loads.
Seven-year depreciation; zero residual	Accounting illustration	Expected service life, replacement plan and local accounting policy.
$12 ex-factory shirt sales	Illustrative selling price	Customer orders for the exact garment and delivery scope.
Optional process pricing	Not established	Module quotes, quality tests, consumables, cycle times and demand.

Procurement and engineering questions

Ask suppliers for exact input/output conditions, control interfaces, fault states, software access, licensing, supported material limits, maintenance intervals and service lead times. Require full costs for necessary air, vacuum, guarding and calibration. Clarify whether advertised cycle time includes loading, cutting, offloading, cleaning and thread changes. Verify that the component can carry its gripper and material at the required reach, not merely its nominal payload.

Ask pilot operators for matched samples, historical order quantities, setup time, reject and rework costs, fabric and trim quotes, actual staffing, floor and utility costs, delivery expectations and willingness to accept disclosed capability limits. This replaces speculative market enthusiasm with a testable process and profit comparison.

Appendix E / Glossary

Term	Meaning
Assembly compiler	Software translating a structured garment into a checked, machine-specific production plan.
Canonical garment record	The approved structured source for geometry, materials, construction, quality and production identity.
Construction family	A group of garments using a related set of qualified operations and tooling.
Cut-and-sew	Construction from cut textile panels joined through the specified sewing operations.
Differential feed	Controlled relative advancement of fabric layers to preserve alignment or an intended length relationship.
Garment topology	How panels, seams, openings and enclosed sections are connected.
GSM	Grams per square meter, a fabric mass-per-area measure; not a complete handling specification.
Handoff contract	Required identity, pose, support, readiness and success conditions when material changes control.
Marker / nesting	Arrangement of pattern components within usable fabric area while respecting constraints.
MOQ / MCQ	Minimum order quantity / minimum color quantity; constraints can arise at different supply-chain stages.
Physical AI	Learned perception, planning or control interacting with the material world; not a safety certification or proof of generality.
Recipe	Versioned, qualified material, geometry, tooling, construction and inspection configuration.
Singulation	Separating the intended individual layer or component from a stack or surrounding material.
Support-and-access map	An operation-by-operation account of how the garment is held, reached and released.
WIP	Work in process: material or garments that have started but are not complete accepted output.

References / Primary-source register

Sources below were reviewed for the 19 October 2026 research snapshot. Public manufacturer pages are attributed descriptions, not independent verification. Research sources are used only for their stated task and preparation conditions. Some sources provide abstracts or public summaries rather than full methodology; their use is correspondingly limited. Standards references concern public scope only. No cited organization is represented as a partner, endorser or licensor of the proposed system.

[01] Lectra. Virga Fashion: on-demand automatic single-ply fabric cutting solution.

Manufacturer documentation: rollout, positioning and cutting; downstream autonomous sewing is outside the advertised cutter function.

Open primary source | Accessed 19 October 2026

[02] RobotFactory. RPT 130 Textile Gripper application.

Vendor application: cutter-linked T-shirt component pickup and sorting; example components then go to a sewing department.

Open primary source | Accessed 19 October 2026

[03] Tang, K., et al. (2025). Fixture-Free Automated Sewing System Using Dual-Arm Manipulator and High-Speed Fabric Edge Detection.

IEEE Robotics and Automation Letters 10(9), 8962–8969. DOI: 10.1109/LRA.2025.3592145. Prepared aligned fabric-pair sewing; public research record and abstract.

Open primary source | Accessed 19 October 2026

[04] Jin, B., et al. (2025). Automated Straight-Line Sewing of Stretchable Fabrics With Different Lengths.

Author manuscript publicly posted July 2026; IEEE Robotics and Automation Letters. DOI: 10.1109/LRA.2025.3619791. Leading ends start under the presser foot; straight-seam experiments.

Open primary source | Accessed 19 October 2026

[05] Kobayashi, A., et al. (2025). RollUP: Rolling-Up End-Effector for Fabric Handling.

IEEE Robotics and Automation Letters 10(11), 11204–11211. DOI: 10.1109/LRA.2025.3609086. Rolling and suction-assisted panel transport; public research record.

Open primary source | Accessed 19 October 2026

[06] Dong, W., et al. (2026). Robotic Fabric Alignment System for Sewing Using Global Local Weighted ICP.

IEEE Transactions on Automation Science and Engineering 23, 13391–13406. DOI: 10.1109/TASE.2026.3713128. CAD-assisted alignment starting with flat, wrinkle-free panels.

Open primary source | Accessed 19 October 2026

[07] Sunil, N., et al. (2022). Visuotactile Affordances for Cloth Manipulation with Local Control.

Conference on Robot Learning; vision and tactile cloth-edge manipulation, not complete garment sewing.

Open primary source | Accessed 19 October 2026

[08] Korosteleva, M., and Sorkine-Hornung, O. (2023). GarmentCode: Programming Parametric Sewing Patterns.

ACM Transactions on Graphics / SIGGRAPH Asia. DOI: 10.1145/3618351. Component-oriented parametric pattern construction.

Open primary source | Accessed 19 October 2026

[09] Zhou, F., et al. (2025). Design2GarmentCode: Turning Design Concepts to Tangible Garments Through Program Synthesis.

CVPR 2025; arXiv:2412.08603. Multimodal design inputs to parameterized garment programs and sewing patterns.

Open primary source | Accessed 19 October 2026

[10] JUKI. JTR-MF7923/BHA bottom-hemming automat.

Manufacturer specifications: an operator-loaded bottom-hemming process. Published capacity is one operation, not complete shirts.

Open primary source | Accessed 19 October 2026

[11] OSHIMA. Fabric preparation and relaxing machines.

Manufacturer explanation of residual fabric tension and material-dependent relaxation before cutting.

Open primary source | Accessed 19 October 2026

[12] Tajima. SAI embroidery machine.

Manufacturer description of eight-needle equipment and frames; automatic hooping and backing handling are not established by this source.

Open primary source | Accessed 19 October 2026

[13] Brother. GTX300 direct-to-garment process and specifications.

Manufacturer requirements for pretreatment and fixing/drying in the selected printing workflow.

Open primary source | Accessed 19 October 2026

[14] SoftWear Automation. Sewbots and on-demand garment worklines.

Company description of machine-vision garment production and sourcing. Public claims are not independently audited here.

Open primary source | Accessed 19 October 2026

[15] Silana. Product: autonomous, local, cost-competitive manufacturing.

Company-described cutting, singulation and adaptive sewing. Broad performance and autonomy claims are not treated as audited results.

Open primary source | Accessed 19 October 2026

[16] CreateMe. Physical AI research and development.

Company-described hardware/control co-design, bonded assembly and generation status; a development roadmap is not a completed capability demonstration.

Open primary source | Accessed 19 October 2026

[17] CreateMe. Brand manufacturing offering.

Company page describing small-batch bonded manufacturing and advertised 100-unit minimum order/color quantities.

Open primary source | Accessed 19 October 2026

[18] Resonance. The technology behind Resonance / CreateOne.

Company account of design-to-production software and manufacturing-network orchestration; not verification of an equivalent standalone autonomous cell.

Open primary source | Accessed 19 October 2026

[19] Sewbo. Robotic garment assembly technology and frequently asked questions.

Temporary fabric stiffening, robotic sewing and treatment removal; public status describes continuing development and partner trials.

Open primary source | Accessed 19 October 2026

[20] SHIMA SEIKI. WHOLEGARMENT.

Manufacturer description of whole-garment knitting from yarn, an adjacent process rather than roll-fed cut-and-sew.

Open primary source | Accessed 19 October 2026

[21] Printful. How to make custom all-over-print shirts.

Service-provider description of selected customization and cut-and-sew products; not unrestricted physical garment automation.

Open primary source | Accessed 19 October 2026

[22] Pine Crest Fabrics. Frequently asked questions.

Supplier-stated stock and made-to-order fabric minimums; not a source for the modeled material prices.

Open primary source | Accessed 19 October 2026

[23] ISO. ISO 10218-2:2025: Robotics — Safety requirements — Part 2: Industrial robot applications and robot cells.

Official public scope only. The complete standard and jurisdiction-specific requirements require qualified engineering review.

Open primary source | Accessed 19 October 2026

[24] ISO. ISO 12100:2010: Safety of machinery — General principles for design — Risk assessment and risk reduction.

Official public scope; this paper does not perform a conformity assessment.

Open primary source | Accessed 19 October 2026

[25] ISO. ISO 6330:2021: Textiles — Domestic washing and drying procedures for textile testing.

Official public scope. Actual garment acceptance methods and tolerances must be specified separately.

Open primary source | Accessed 19 October 2026

[26] NIST. AI Risk Management Framework.

Voluntary AI-risk governance reference, not industrial-machinery safety certification.

Open primary source | Accessed 19 October 2026

[27] United Nations Environment Programme (2023). Sustainability and Circularity in the Textile Value Chain: A Global Roadmap.

Official roadmap summary; used for the value-chain evaluation boundary, not for a quantified environmental benefit of this system.

Open primary source | Accessed 19 October 2026

[28] UFACTORY. Lite 6 collaborative robot.

Manufacturer-advertised starting price, payload and reach; illustrative warning against comparing robot prices without their operating limits.

Open primary source | Accessed 19 October 2026

[29] Chen, D., et al. (2026). Phase-Conditioned Imitation Learning with Autonomous Failure Recovery for Robust Deformable Object Manipulation.

Author research report on T-shirt hanging and removal; not autonomous seam repair or complete production-cell recovery.

Open primary source | Accessed 19 October 2026